

EDUCATIVE COMMENTARY ON ISI 2025 MATHEMATICS PAPERS

(Updated on 31/03/2026)

Introduction	1
Paper 1	3
Paper 2	41
Concluding Remarks	76
Appendix	84

Introduction

The original motivation for writing the annual commentaries on ISI B.Stat. Entrance Tests is now waning. They began in 2016 because it was the year in which some exciting questions came to the notice of the author. Later on commentaries from 2014 were added. Unfortunately, after 2017, there was a noticeable decline in the quality of questions. Some readers also brought to the notice of the author that a few questions which were hailed in the commentary were picked from some reputed international contests. This did not detract from the intrinsic quality of the questions. Nor is it likely that some candidates had seen the solutions already. But apart from loss of originality, it is an insult to a good problem (for which originally ample time was allowed) to expect the candidates to crack it in a couple of minutes.

Still, the annual commentaries on ISI Entrance Test continued more as a mission, because some students pointed out that unlike the JEE, there are no sources to get the answers of ISI Entrance tests. But things are changing here too. In addition to the official key to the multiple choice questions in Paper 1, videos giving worked out solutions are available. As for the second paper, solutions for some of the problems are presented on the [Artofsolvingproblems](#).

But even with both the justifications for the commentaries losing their strengths, I believe there is still some room to bring out the educative aspects of some of the problems and solutions. So, here is the commentary on ISI

2025.

I thank Prashant Sharma for sharing his solutions to the problems in Paper 1. The solutions given here are often edited and elaborated versions of the solutions given by him with some comments by me. The major idea behind the alternate solution to Q. 6 of Paper 2 is due to Agnibesh. The link given in the answer to Q. 7 of Paper 2 was provided by Abhishek Bansal who also helped in conversion of files. The short cut to Q.24 of Paper 1 was pointed out by Tech Sanku. But, above all, I thank the readers of the previous commentaries who eagerly awaited the present commentary. Their encouraging feedback keeps me going.

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, Third Edition, published by Universities Press, Hyderabad.

Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2024 Paper 1 with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 Let A be an $m \times n$ matrix with the (i, j) -th entry given by the real number a_{ij} , $1 \leq i \leq m, 1 \leq j \leq n$. Let

$$\alpha = \max_{1 \leq j \leq n} \{ \min_{1 \leq i \leq m} a_{ij} \} \quad \text{and} \quad \beta = \min_{1 \leq j \leq n} \{ \max_{1 \leq i \leq m} a_{ij} \}.$$

Then

- (A) $\alpha \leq \beta$ but not necessarily $\alpha = \beta$
- (B) $\beta \leq \alpha$ but not necessarily $\alpha = \beta$
- (C) $\alpha = \beta$
- (D) nothing can be said in general

Answer and Comments: (D). It is important to understand what α and β really are. In the definition of α , we are first taking the minimum in the j -th column, for every $j = 1, 2, \dots, n$. Then we are taking the maximum of these minima. In β this is reversed, that is we are taking the minimum of the columnwise maxima. If we were taking the maximum of the maximum, we would get the largest entry of the entire matrix. This would be uninteresting. Similarly, the minimum of the columnwise minima would be the smallest entry of the matrix.

The problem becomes interesting because we are comparing the maximum of the columnwise minima with the minimum of the columnwise maxima. However, there is no formula for either α or β , save their definitions. Also a matrix is a function of two discrete variables, i and j . Hence methods of calculus are useless.

In absence of any such helpful tools, it often pays to take a real life model and think in terms of it. For example, suppose there are m cricket players $P_1, P_2, \dots, P_m$ who play in matches $T_1, T_2, \dots, T_n$ of some series. Let a_{ij} be the score of P_i in the match T_j . Then the j -th column is the score board for the T_j -th match. Suppose some player, say P_1 is a poor batsman and hits a duck in every match. Then

because of him, the column minimum in every column will be 0 and hence so will be α , their maximum. So, even without computing β we know that $\alpha \leq \beta$. On the other hand, suppose that for some match, say T_1 , the wicket is so bad that everyone scores a duck. Then for the first column, the maximum is 0. Hence β , which is the minimum of all column maxima would also be 0. This time we have $\beta < \alpha$.

In both the examples, all entries are non-negative. In the first example, the first row is identically 0, in the second example, the first column is identically 0. For concrete examples, one can take $A = \begin{bmatrix} 0 & 0 \\ 100 & 100 \end{bmatrix}$.

Then $\alpha < \beta$. For the second example, let $A = \begin{bmatrix} 0 & 100 \\ 0 & 100 \end{bmatrix}$. Here $\beta < \alpha$. So nothing can be concluded about the relationship between α and β .

■

A simple, but good problem which is superficially about matrices but is really about inequalities.

Of course, the counterexamples could as well have been thought of without the motivation. But the die hard fans of cricket are not likely to buy that.

- Q.2 The coefficient of x^8 in $(1 - 3x)^6 (1 + 9x^2)^6 (1 + 3x)^6$ is -
 (A) $-3^9 \times 5$
 (B) $3^9 \times 5$
 (C) $-3^8 \times 5$
 (D) $3^8 \times 5$

Solution and Comments: (B). Multiplying the first and the last factor and further simplification reduces the given expression, say E as

$$\begin{aligned} E &= (1 - 9x^2)^6 (1 + 9x^2)^6 \\ &= (1 - 81x^4)^6 \end{aligned} \tag{1}$$

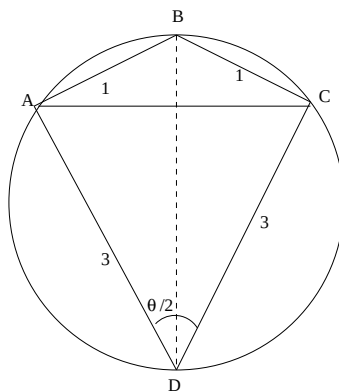
$$= (1 - 81u)^6 \tag{2}$$

where in the last step we have put $u = x^4$. The problem now reduces to finding the coefficient of u^2 in $(1 - 81u)^6$. A direct application of

the binomial theorem gives the answer as $\binom{6}{2}(81)^2$ which simplifies to $3^{18} \times 15 = 3^9 \times 5$. ■

Absolutely simple problem on binomial theorem.

Q.3 Consider the cyclic quadrilateral $ABCD$ given below.



Assume that $AB = BC$, $AD = CD$, and $\frac{AB}{AD} = \frac{1}{3}$. Let $\theta = \angle ADC$. Then $\cos \theta$ is equal to

- (A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{4}{5}$

Answer and Comments: (D). From the data, both B and D lie on the perpendicular bisector of the segment AC which is also a chord of the circle. Hence the triangles BAD and BCD are right angled and congruent to each other. Also $\angle BDA = \theta/2$.

Without loss of generality, assume AB is of 1 unit. Then $AD = 3$ and $BD = \sqrt{9+1} = \sqrt{10}$. Therefore $\cos \frac{\theta}{2} = \frac{3}{\sqrt{10}}$. So, $\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1 = \frac{9}{5} - 1 = \frac{4}{5}$. ■

A simple problem especially because of the symmetry of the data.

Q.4 In the xy -plane, the curve $3x^3y + 6xy + 2xy^3 = 0$ represents

- (A) a pair of straight lines
 (B) an ellipse
 (C) a pair of straight lines and an ellipse
 (D) a hyperbola

Answer and Comments: (A). The given equation is of the fourth degree in x and y . There is a classification of curves whose equation is of degree 1 or 2. But no similar classification in higher degree. However, we see that the expression equated to 0 factors. Hence the curve $3x^3y + 6xy + xy^3 = 0$ is the same as the curve $xy[3x^2 + 2 + y^2] = 0$. Therefore either $xy = 0$ or $3x^2 + 2 + y^2 = 0$. But the second factor is always positive and hence can never vanish. Therefore we must have $xy = 0$ which implies $x = 0$ or $y = 0$. This is a pair of straight lines. ■

A simple problem once the idea of factorisation strikes.

Q.5 Let $I = \int_3^5 \frac{1}{1+x^3} dx$. Then

- (A) $I < \frac{1}{64}$
- (B) $I > \frac{1}{13}$
- (C) $\frac{1}{63} < I < \frac{1}{14}$
- (D) $I > \frac{1}{2} \left(\frac{1}{14} + \frac{1}{63} \right)$

Answer and Comments: (C). The direct approach is to evaluate the integral I and then see which inequalities it satisfies. But finding an antiderivative of $\frac{1}{1+x^3}$ is rather complicated because first you have to factor the denominator, then resolve the integrand into partial fractions and find an antiderivative of each term. The very fact all options give only some upper and lower bounds on I indicates that there has to be some easier way, without evaluating I exactly. And indeed there is one. Suppose m and M are the minimum and the maximum of a continuous function $f(x)$ on an interval $[a, b]$, then we have

$$m \leq f(x) \leq M \quad (1)$$

for all $x \in [a, b]$. Integrals preserve inequalities. So

$$\int_a^b m dx = \int_a^b f(x) dx \leq \int_a^b M dx \quad (2)$$

Hence

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a) \quad (3)$$

If $f(x)$ is monotonically decreasing, there is a bonus because now m and M are simply the values $f(b)$ and $f(a)$ respectively without any further

calculations. In the present problem, $f(x) = \frac{1}{1+x^3}$. The denominator is strictly increasing and hence its reciprocal is strictly decreasing. (This can also be seen by finding $f'(x)$. But why unnecessarily spend time on it?) So $M = f(a) = f(3) = \frac{1}{28}$ and $m = f(b) = f(5) = \frac{1}{126}$. Putting these into (3),

$$\frac{1}{63} < I < \frac{1}{14} \quad (4)$$

which is (C). ■

The bounds we derived for I are too crude. Suppose c is the midpoint of $[a, b]$, i.e. $c = \frac{a+b}{2}$. Then the intervals $[a, c]$ and $c, b]$ are both of half the length of $[a, b]$. We can split the integral I into two parts $I_1 + I_2$ where $I_1 = \int_a^c f(x)dx$ and $I_2 = \int_c^b f(x)dx$. Note that $f(c)$ now is simultaneously the minimum of $f(x)$ on $[a, c]$ and also its maximum on $[c, b]$. Applying (3) separately to I_1 and I_2 , we get

$$\begin{aligned} f(c)\frac{b-a}{2} &< I_1 < f(a)\frac{b-a}{2} \\ \text{and } f(b)\frac{b-a}{2} &< I_2 < f(c)\frac{b-a}{2} \end{aligned} \quad (5)$$

Adding,

$$\frac{f(c) + f(b)}{2}(b-a) < I < \frac{f(c) + f(a)}{2}(b-a) \quad (6)$$

In the present problem, $a = 3, b = 5$ and $c = 4$. $f(4) = \frac{1}{65}$. This gives

$$\frac{1}{65} + \frac{1}{126} < I < \frac{1}{65} + \frac{1}{28} \quad (7)$$

which is predictably sharper than (C), because the lower bound is bigger and the upper bound smaller than that in (4). The problem would have been a little more challenging had this inequality been given as an option. And even more challenging if (7) were given in the form

$$\frac{191}{8190} < I < \frac{93}{1820} \quad (8)$$

It would take unusual perceptivity to recognise that 65 is a common factor of the denominators and to guess what possible role $\frac{1}{65}$ has in this problem.

We can, of course, add more points of subdivision and get better bounds on I . As the mesh of the subdivision tends to 0, both the bounds tend to the exact value of I , by very definition of Riemann integral. When the nature of the integrand is such that finding its antiderivative is impossible or very difficult to calculate exactly, this method which demands only arithmetic can be used to approximate it to a desired degree of accuracy. This approximation to an integral is called its **trapezoidal approximation** because it consists of dividing the region below the graph of $y = f(x)$ approximately into vertical trapezoids.

Q.6 Consider two events A and B with probabilities $P(A)$ and $P(B)$ respectively such that $0 < P(A), P(B) < 1$. Define

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Consider the following statements.

$$(I) P(A | B') + P(A | B) = 1.$$

$$(II) P(A' | B) + P(A | B) = 1 ?$$

Then in general,

(A) (I) is true and (II) is false

(B) (I) is false and (II) is true

(C) Both (I) and (II) are true

(D) Both (I) and (II) are false

Answer and Comments. (B) Those who know Bayes law will recognise that what is given as $P(A|B)$ is simply the conditional probability of A given B . However, even without this knowledge, we can mechanically calculate the expressions appearing in the two statements.

For (I), we have

$$\begin{aligned} P(A|B') + P(A|B) &= \frac{P(A \cap B')}{P(B')} + \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A) - P(A \cap B)}{1 - P(B)} + \frac{P(A \cap B)}{P(B)} \end{aligned}$$

$$= \frac{P(A)P(B) - 2P(B)P(A \cap B) + P(A \cap B)}{P(B)(1 - P(B))}$$

Nothing is given about the mutual relationship between the events A and B . They could as well be mutually exclusive. Then $P(A \cap B) = 0$ and the expression reduces to $\frac{P(A)P(B)}{P(B)(1-P(B))}$ which equals $\frac{P(A)}{1-P(B)}$. This will equal 1 if and only if $P(A) + P(B) = 1$ but not in general. (As a concrete example, suppose a ball is drawn at random from an urn containing 1 red, 1 blue and 1 green ball and A, B are the events that it is red and it is blue respectively. Here $P(A) = P(B) = \frac{1}{3}$ and A, B are mutually exclusive.)

However, for (II) we have

$$\begin{aligned} P(A'|B) + P(A|B) &= \frac{P(A' \cap B)}{P(B)} + \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(B) - P(A \cap B) + P(A \cap B)}{P(B)} \\ &= 1 \end{aligned} \tag{1}$$

Therefore (I) is false and (II) is true. ■

Even without these elaborate calculations, the truth of II can be seen as follows. $P(A'|B)$ is the conditional probability that A' occurs (i.e. A fails) given that B does. Similarly for $P(A|B)$ is the probability that A occurs given that B occurs. Now, when B occurs, exactly one of A' and A occurs. So, the two probabilities $P(A|B)$ and $P(A'|B)$ always add to 1. Hence (II) is always true. But in (I), even if the occurrence of B forces A to occur with probability $P(A|B)$, nothing can be said in general about the extent to which the non-occurrence of B forces the occurrence of A . (As a real life example, if a rich person has probability $\frac{1}{3}$ of being happy, we can say absolutely nothing about the probability of a poor person being happy.) Hence (I) is not true in general. In probability problems, sometimes common sense works faster than formulas.

Q.7 Let $f(x) = 7x^{11} + 4x^5 - 3$. Then f has

- (A) exactly 1 real root
- (B) exactly 3 real roots

- (C) exactly 5 real roots
 (D) 11 real roots

Answer and Comments: (A). Being an odd degree polynomial, $f(x)$ has at least one real root. If it has some other root, then by Rolle's theorem, the derivative $f'(x)$ will vanish at some point between these two roots. By a direct calculation, $f'(x) = 77x^{10} + 20x^4$. This is positive except at 0. But even this is enough to show that f is strictly increasing on the entire real line. Let $a, b \in \mathbb{R}$ with $a < b$. By Lagrange's MVT, there is some $c \in (a, b)$ such that $f(b) - f(a) = f'(c)(b - a)$. If both a, b are positive, so is c . If both a, b are negative then so is c . In either case, $f'(c) > 0$ and we get $f(b) - f(a) > f'(c)(b - a) > 0$. The only remaining case is when $a < 0$ but $b > 0$. In this case we apply LMVT separately to the intervals $[0, b]$ and $[a, 0]$. Note that the point c given by LMVT always lies in the open intervals $(0, b)$ and $(a, 0)$ as the case may be. Hence $f'(c) > 0$. So, we get $f(a) < f(0) < f(b)$. (More generally, this argument shows that if $f'(x) \geq 0$ everywhere except possibly at a finite number of points in (a, b) , then $f(a) < f(b)$. This is how one shows that the function $\sin x + x$ is strictly increasing on $\mathbb{R}$.)

Coming back to our problem, $f(x)$ is strictly increasing on $\mathbb{R}$. Hence it cannot have more than one real root. We have already shown that it has at least one root. Put together, (A) holds. ■

A simple problem, but not suitable as an MCQ as it does not test the ability to show that sometimes a function is strictly increasing even when its derivative is non-negative everywhere but vanishes at a few points.

- Q.8 Twelve boxes are placed along a circle. In each box, 1,2,3 or 4 balls are put such that the total number of balls in any 4 consecutive boxes is same. The number of ways this can be done is

- (A) $4!$ (B) 4^4 (C) $(4!)^3$ (D) $(4!)^4$

Answer and Comments: (B). Number the boxes in a clockwise order as $b_1, b_2, \dots, b_{12}$. Let x_i be the number of balls in b_i for $1 \leq i \leq 12$. Each x_i can take four possible values. But the requirement given means in particular, that $x_1 + x_2 + x_3 + x_4 = x_2 + x_3 + x_4 + x_5$. Hence $x_1 = x_5$.

With the same reasoning, $x_5 = x_9$. Similarly, $x_2 = x_6$ etc. Hence the boxes fall into 4 groups $G_1 = \{b_1, b_5, b_9\}$, $G_2 = \{b_2, b_6, b_{10}\}$, $G_3 = \{b_3, b_7, b_{11}\}$ and $G_4 = \{b_4, b_8, b_{12}\}$. The same number of balls must be put in all boxes of the same group. But these numbers can be picked independently from the set $\{1, 2, 3, 4\}$. As there are four groups and 4 free choices for each group, the number of possible placements is 4^4 . ■

A simple but good problem on combinatorics. The crucial point is to translate the condition given in terms of the formation of four groups each containing three boxes. After that it is easy. Although the reasoning takes long to write, once it is conceived, the calculations are minimal. Hence the problem is suitable as an MCQ. But it fits even more properly into fill into the blanks type problems where only the answer and no reasoning needs to be given. The moment 4^4 is seen as an answer, it is safe to presume that the candidate has done the correct reasoning. This is different from the last problem, where there is no way to tell merely from the answer whether the candidate has correctly handled the case where the derivative is non-negative everywhere but vanishes at a single point.

Q.9 Let $A = \{(x, y) : x, y \in [0, 1]\}$ and $B = \{(x, y) : x, y \in [0, 2]\}$. Define $f : A \rightarrow B$ by $f(x, y) = (x^2 + y, x + y^2)$.

Then f is

- (A) one-to-one but not onto
- (B) onto but not one-to-one
- (C) both one-to-one and onto
- (D) neither one-to-one nor onto

Answer and Comments: (D). When there is some characterisation of some property, there is often a tendency to test that property by testing the characterisation rather than testing the original property. For example, when it comes to checking whether a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is monotonically increasing or decreasing. The first thing which many people do almost automatically is to consider its derivative f' and check its sign. Same for the second derivative test for local maximum/minimum. As happened in Q.5 above, checking that the integrand is monotonically decreasing directly from the definition was

much easier.

Fortunately this temptation is nipped in the bud when it comes to checking if a given function is one-to-one or onto. There is indeed a characterisation in terms of the existence of a left/right inverse. But that is not easy to apply. So the best thing to do is to check these properties directly for the given function $f : A \longrightarrow B$.

For f to be one-to-one, we must show that if $f(x_1, y_1) = f(x_2, y_2)$ for any $(x_1, y_1), (x_2, y_2) \in A$, then we must have $x_1 = x_2$ and $y_1 = y_2$. In the present problem, the

$$f(x_1, y_1) = (x_2, y_2) \tag{1}$$

means two simultaneous equations

$$x_1^2 + y_1 = x_2^2 + y_2 \tag{2}$$

$$\text{and } x_1 + y_1^2 = x_2 + y_2^2 \tag{3}$$

It is impossible to solve this system as there are only two equations but four variables. However, we can check the two equalities separately. In (2), it may happen that $x_1 < x_2$ in which case $x_1^2 < x_2^2$. But the gap between them is at most 1 since both x_1, x_2 vary only in $[0, 1]$. This difference can be made up by choosing $y_1 > y_2$ suitably. This will not harm (3) because $y_1^2 < y_2^2$ whenever $0 \leq y_1 < y_2 \leq 1$. The maximum gap arises when $x_1 = 0$ and $x_2 = 1$. In that case, (2) forces $y_1 = 1$ and $y_2 = 0$. Luckily, (3) also holds with these values. Hence $f(1, 0)$ and $f(0, 1)$ both equal $(1, 1)$. Therefore f is not one-to-one.

To check whether f is onto, note that the codomain B of f is the square $[0, 2] \times [0, 2]$. For f to be onto, for every point, say (a, b) , in this square, we must be able to find some $x, y \in [0, 1]$ such that

$$x^2 + y = a \tag{4}$$

$$\text{and } x + y^2 = b \tag{5}$$

Solving this system is not easy because if we eliminate y , we get a quartic in x . Hence even if f is onto, proving it is so is not easy. Of course, this by itself does not prove that f is *not* onto. But it does suggest that we should first try to show that it is not onto. For this, it suffices to find even one point in the codomain B which has no preimage

in A . The first try is the four corners of the square $[0, 2] \times [0, 2]$. The corners $(0, 0)$ and $(2, 2)$ are covered by the points $(0, 0)$ and $(1, 1)$ of A respectively. But what about the other two corners $(2, 0)$ and $(0, 2)$? If there is some $(x, y) \in A$ for which $f(x, y) = (2, 0)$, then the system above must have a solution with $a = 2$ and $b = 0$. But since $0 \leq x \leq 1$ and $0 \leq y \leq 1$, (4) with $a = 2$ forces $x = 1, y = 1$. In that case (5) cannot hold with $b = 0$.

Thus we have shown that there is at least one point of the codomain, viz. $(2, 0)$ which has no preimage under f . Therefore f is not onto. As it is not one-to-one either, (D) holds. ■

A good problem on testing some elementary set theoretic properties of functions. Proving that f is not one-to-one is easy. Proving that it is not onto is also not difficult after sensing the difficulty in proving that it is onto. Some persons can get the right inkling for a wrong reason, that is the area of the codomain B is much greater than that of the domain A . But this is a wrong reasoning as proved by the function $g(x, y) = (2x, 2y)$ which is onto.

- Q.10 The number of ordered pair (a, b) of positive integers with $a < b$ satisfying $a^2 + b^2 = 2025$ is
- (A) 0
 - (B) 1
 - (C) 2
 - (D) 6

Answer and Comments: (B). The first thing to note is that 2025 is itself a perfect square, viz. $(45)^2$. So the problem asks to find all triples (a, b, c) in which $a^2 + b^2 = c^2$ with $c = 45$. (For the time being, we ignore the requirement $a < b$ as it is merely to ensure that no triple is counted twice.) For obvious reasons, such a triple is called a **Pythagorean triple**. Evidently, if (a, b, c) is a Pythagorean triple, then so is (ka, kb, kc) for any positive integer k . We may say that it is an ‘inflated’ triple. Conversely, if d is a common divisor of a, b, c , then $(\frac{a}{d}, \frac{b}{d}, \frac{c}{d})$ is also a Pythagorean triple (a ‘deflated’ triple).

There is a way to construct all Pythagorean triples. For any positive integers u, v with $u > v$, it is seen by a direct computation that $(u^2 -$

$v^2, 2uv, u^2 + v^2$) is a Pythagorean triple. The simplest Pythagorean triple is $(3, 4, 5)$ which arises by taking $u = 2$ and $v = 1$. But triples of this form are not the only Pythagorean triples. The inflated triple $(6, 8, 10)$ is a Pythagorean triple which can be obtained in this way by taking $u = 3, v = 1$. However, the Pythagorean triple $(9, 12, 15)$ cannot be obtained this way because there is no way to express 15 as a sum of two perfect squares $u^2 + v^2$. (To see this, the possible candidates for u are 1, 2 and 3. But none of $15 - 1, 15 - 4$ and $15 - 9$ is a perfect square.)

$(3k, 4k, 5k)$ is a Pythagorean triple for every positive integer k . If we take $k = 9$, we get $(27, 36, 45)$ is a Pythagorean triple of the desired type. Hence there is at least one such triple.

We claim that it is the only such triple. For this we need a result from Exercise (4.22)(b) which says that a Pythagorean triple (a, b, c) in which every two of the entries are relatively prime comes from a unique pair u, v . Such a triple is often called a **primitive** Pythagorean triple. Because of this result, we can start with any Pythagorean triple (a, b, c) , and take the g.c.d., say d , of a, b, c . Then $(\frac{a}{d}, \frac{b}{d}, \frac{c}{d})$ is a triple which is of the form $(u^2 - v^2, 2uv, u^2 + v^2)$ for some integers u, v . This triple may be called the reduction of the original Pythagorean triple. Its advantage is that it can always be constructed from some suitable u, v . Also we can recover the original triple (a, b, c) from it, by inflating it with the factor d .

We apply this to test how many Pythagorean triples of the form $(a, b, 45)$ there are. Given any such triple, let $(\frac{a}{d}, \frac{b}{d}, \frac{45}{d})$ be its reduced Pythagorean triple where d is the g.c.d. of a, b and 45. We do not know a, b . Hence we do not know d . But it is definitely a divisor of 45. Since $45 = 3^2 \times 5$, the possible values of d are 1, 3, 5, 9 and 15. (We reject 45 since it does not divide a and b each of which is less than 45.)

We settle these cases one by one.

- (i) $d = 1$. Then $\frac{45}{d} = 45$. 45 can be expressed as $u^2 + v^2$ with $u > v$ only in one way, viz. with $u = 6, v = 3$. This does give $(27, 36, 45)$ as a Pythagorean triple of the desired type.
- (ii) $d = 3$. Here $\frac{45}{3} = 15$. As already proved, 15 cannot be expressed as a sum of two squares. Hence there is no primitive

Pythagorean triple in which the last entry is 15. (There is the Pythagorean triple (9, 12, 15). But it is not primitive. Its inflated triple (27, 36, 45) is already counted.)

- (iii) $d = 5$. Here $\frac{45}{d} = 9$. Here also, 9 cannot be expressed as a sum of two squares and hence there is no primitive triple whose last entry is 9. Therefore there is no Pythagorean triple $(a, b, 45)$ whose reduction has 9 as its last entry.
- (iv) $d = 9$. This time $\frac{45}{d} = 5$ which can indeed be expressed as $u^2 + v^2$ with $u = 2, v = 1$. The resulting primitive Pythagorean triple is (3, 4, 5). Inflating it by a factor of 9, we get (27, 36, 45) which is already counted.
- (v) Finally, suppose $d = 15$. Then $\frac{45}{d} = 3$. Since 3 cannot be expressed as a sum of two squares, this case is similar to (iii) and gives rise to no triple $(a, b, 45)$ of the desired type.

After exhausting all possibilities, we see that there is only one pair (27, 36) of the desired type. ■

The essential cause of the difference between cases (iii) and (iv) is a basic difference between primes of the form $4k + 1$ and those of the form $4k + 3$. The former can be expressed as sums of two squares. But the latter cannot. Based on this difference, there is a criterion to tell in how many ways a given integer n can be expressed as a sum of two squares, by looking at its prime power factorisation and counting how many primes of each type there are and with what multiplicities. Although this criterion is not difficult to state, its proof requires good delving into what is called the ring of **Gaussian integers**. (These are complex numbers whose real as well as imaginary parts are integers, sometimes called the lattice points.) So we have avoided that approach.

The problem is yet another casualty of the MCQ format. It is impossible to distinguish between a candidate who answers it correctly by analysing all possibilities and concluding that only one is successful from another who can identify one successful instance by inspection and simply can't think of any others. The situation could have been salvaged a little by giving, instead of 45, a number such as 65 which can be expressed as a sum of two squares in two different ways ($64 + 1$ or $49 + 16$).

Q.11 For each $n \geq 1$, let a_n and b_n be real numbers such that $a_n \neq 0$ and $b_n \neq 0$. Let $(a_n + ib_n)^n = n(a_n + ib_n)$ for all $n \geq 6$.

Then

(A) no such a_n, b_n exist

(B) $a_n = n^{\frac{1}{n+1}} \cos \frac{2\pi}{n+1}, b_n = n^{\frac{1}{n+1}} \sin \frac{2\pi}{n+1}$

(C) $a_n = n^{\frac{1}{n}} \cos \frac{2\pi}{n}, b_n = n^{\frac{1}{n}} \sin \frac{2\pi}{n}$

(D) $a_n = n^{\frac{1}{n-1}} \cos \frac{2\pi}{n-1}, b_n = n^{\frac{1}{n-1}} \sin \frac{2\pi}{n-1}$

Answer and Comments: (D). The problem is not about any sequence. n is a fixed positive integer. So, no harm in calling $a_n + ib_n$ simply as z , rather than z_n . This is a non-zero complex number and we are given that

$$z^n = nz \tag{1}$$

As $z \neq 0$, this implies

$$z^{n-1} = n \tag{2}$$

Hence z is an $(n-1)$ -th root of the complex number n . In all every non-zero complex number has $n-1$ distinct complex $(n-1)$ -th roots. When the number is a positive real number, they consist of the real positive $(n-1)$ -th root of it multiplied by the $n-1$ $(n-1)$ -th roots of unity. The latter are numbers of the form $e^{\frac{2\pi ik}{n-1}}$, for $k = 1, 2, \dots, n-1$. Hence there are $n-1$ solutions of (1), one of which (obtained by taking $k = 1$) is $a_n + ib_n$ where

$$a_n = n^{\frac{1}{n-1}} \cos \frac{2\pi}{n-1} \quad \text{and} \quad b_n = n^{\frac{1}{n-1}} \sin \frac{2\pi}{n-1} \tag{3}$$

Hence out of the given options (D) is correct. ■

The problem is ill-framed. It is not clear what role is played by the stipulation $n \geq 6$. (One can understand exclusion of $n = 1$ because $n-1$ appears in the denominator $n-1$ of the exponent. But values $n \geq 2$ are all right.) Moreover, there are $n-2$ other solutions besides those given by (D). The requirements that a_n, b_n are both non-zero will preclude some more values. Still many are left. So, the data does not conclusively imply (D).

Q.12 Let $a_0 = 0, a_1 = 1$ and $a_n = 5a_{n-1} + a_{n-2}$ for $n \geq 2$. Then the value of the determinant $\begin{vmatrix} a_{101} & a_{100} \\ a_{100} & a_{99} \end{vmatrix}$ is

- (A) -1 (B) -5^{101} (C) 1 (D) 5^{101}

Answer and Comments: (C). A typical example of a sequence defined by a linear recurrence relation of order 2. (The most famous example is, of course, the Fibonacci sequence.) There is a standard procedure to express a_n in terms of the powers of the numbers α and β , defined as the roots of the characteristic equation, which in the present problem is the quadratic $x^2 - 5x - 1 = 0$. Once that is obtained any question about the sequence can be answered. But when these roots are irrational, the procedure becomes cumbersome. So, one looks for short cuts tailored for a particular part of the problem. The present problem asks for the value of $a_{101}a_{99} - a_{100}^2$. Let us, more generally consider $a_{n+1}a_{n-1} - a_n^2$ for any n . The paper-setters have given an implicit hint by giving this expression as a determinant, for then some properties of determinants could be useful.

Once this idea strikes, call the value of the determinant as Δ_n . Substituting the relation for the entries in the first column, we get

$$\begin{aligned} \Delta_n &= \begin{vmatrix} a_{n+1} & a_n \\ a_n & a_{n-1} \end{vmatrix} \\ &= \begin{vmatrix} 5a_n + a_{n-1} & a_n \\ 5a_{n-1} + a_{n-2} & a_{n-1} \end{vmatrix} \end{aligned} \quad (1)$$

$$= \begin{vmatrix} 5a_n & a_n \\ 5a_{n-1} & a_{n-1} \end{vmatrix} + \begin{vmatrix} a_{n-1} & a_n \\ a_{n-2} & a_{n-1} \end{vmatrix} \quad (2)$$

where in going from (1) to (2), we have split the first column. In (2), the first determinant vanishes since its first column is a multiple of the second. After interchanging the columns, the second determinant is precisely Δ_{n-1} . Hence

$$\Delta_n = -\Delta_{n-1} \quad (3)$$

Repeated applications give $\Delta_n = -\Delta_1$ for even n . By direct substitution, $\Delta_1 = \begin{vmatrix} a_2 & a_1 \\ a_1 & a_0 \end{vmatrix} = \begin{vmatrix} 5 & 1 \\ 1 & 0 \end{vmatrix} = -1$. So $\Delta_{100} = 1$. ■

A very well designed problem duly rewarding those who focus on the essentials, unduly rewarding those who venture a guess from the values of Δ_n for $n = 1, 2, 3$ and penalising those who waste their time in solving the recurrence relation to get a closed form expression for a_n .

- Q.13 The lengths of the three sides of a right angled triangle are in geometric progression. The smallest angle of the triangle is
- (A) $\tan^{-1} \left(\frac{\sqrt{5}-1}{2} \right)$
 - (B) $\cos^{-1} \left(\frac{2}{\sqrt{5}-1} \right)$
 - (C) $\sin^{-1} \left(\frac{2}{\sqrt{5}-1} \right)$
 - (D) $\sin^{-1} \left(\frac{\sqrt{5}-1}{2} \right)$

Answer and Comments: (D). The largest angle of a right angled triangle is $\frac{\pi}{2}$. Let α be the smallest angle. Then the third angle $\frac{\pi}{2} - \alpha$ lies in between the two. That is

$$\alpha \leq \frac{\pi}{2} - \alpha < \frac{\pi}{2} \quad (1)$$

Since the sine function is strictly increasing in the first quadrant, we have

$$\sin \alpha \leq \cos \alpha < 1 \quad (2)$$

The sides are not given. But they are given to be in G.P. Hence so are the sines of the angles as they are proportional to the sides. In any G.P. of length 3, the square of the middle term is the product of the other two terms. So we get

$$\cos^2 \alpha = \sin \alpha \quad (3)$$

This gives a quadratic in $\sin \alpha$, viz.

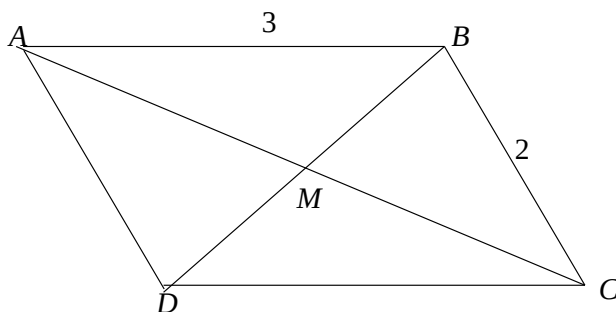
$$\sin^2 \alpha + \sin \alpha - 1 = 0 \quad (4)$$

Solving, $\sin \alpha = \frac{-1 \pm \sqrt{5}}{2}$. We discard the negative sign since $\sin \alpha > 0$. So $\sin \alpha = \frac{\sqrt{5}-1}{2}$. As α is an acute angle it follows that (D) holds. ■

A somewhat unusual, but cute problem about right angled triangles.

- Q.14 The lengths of the two adjacent sides of a parallelogram are 2 cm and 3 cm . The length of one diagonal is $\sqrt{19}$. Then the length of the other diagonal is
- (A) $\sqrt{5}$ cm
 (B) $\sqrt{7}$ cm
 (C) $\sqrt{15}$ cm
 (D) $\sqrt{21}$ cm

Answer and Comments: (B). Suppose $ABCD$ is a parallelogram with $AB = 3$ and $BC = 2$. (All measurements are in centimeters.) Let AC be the diagonal with length $\sqrt{19}$. Suppose that the two diagonals intersect at M . Then M is the midpoint of both the diagonals.



Suppose that the diagonal AC has length $\sqrt{19}$. Then $AM = \frac{\sqrt{19}}{2}$. We apply Appolonius theorem to the triangle ABC and get

$$\begin{aligned} 9 + 4 &= 2(MB^2 + MA^2) \\ &= 2(MB^2 + \frac{19}{4}) \end{aligned} \tag{1}$$

This gives

$$MB^2 = \frac{13}{2} - \frac{19}{4} = \frac{26 - 19}{4} = \frac{7}{4} \tag{2}$$

Hence $MB = \frac{\sqrt{7}}{2}$. But this is half the diagonal BD . Therefore $BD = \sqrt{7}$. ■

An absolutely simple problem in geometry once the right theorem is applied.

Q.15 Consider the following statements about two similar triangles Δ_1 and Δ_2 .

s_1 : Lengths of the sides of Δ_1 are in arithmetic progression.

s_2 : Lengths of the sides of Δ_1 are in geometric progression.

s_3 : Lengths of the sides of Δ_2 are in arithmetic progression.

s_4 : Lengths of the sides of Δ_2 are in geometric progression.

Then

(A) s_1 implies s_3 , but s_2 does not imply s_4

(B) s_1 does not imply s_3 , but s_2 implies s_4

(C) s_1 implies s_3 , and s_2 implies s_4

(D) s_1 does not imply s_3 , and s_2 does not imply s_4

Answer and Comments: (C). This is more a question about progressions than geometry. The only property needed about similar triangles is that their corresponding sides are proportional. So assume that ABC and $A'B'C'$ are similar. As usual denote their sides by a, b, c and a', b', c' . Then there is some (positive) constant k such that

$$a' = ka, b' = kb \text{ and } c' = kc \quad (1)$$

Assume for the sake of definiteness that $a \leq b \leq c$. Now we tackle the options one by one.

(A) If s_1 holds, then $2b = a + c$. But then because of (1), $2kb' = ka' + kc'$. Hence s_1 implies s_3 . However, if s_2 holds, then we have $b^2 = ac$. For a', b', c' to be in G.P. we must have $b'^2 = a'c'$, i.e. $k^2b^2 = kakc$ which is true. Hence s_2 implies s_4 . So (A) is false.

(B) We already showed that s_1 implies s_3 . That alone renders (B) false.

(C) We already showed that s_1 implies s_3 and s_2 implies s_4 . So (C) is true.

(D) We need not check this as only one option is true. Nevertheless, for the sake of completeness, (D) is false, because its first part, viz. s_1 does not imply s_3 is false.

Summing up. option (C) is true. ■

Absolutely simple problem requiring only a property of similar triangles and little more than the definitions of the progressions.

Q.16 Let $[x]$ denote the greatest integer less than or equal to $x \in R$ and $|x|$ has its usual meaning, that is, $|x| = x$ if $x \geq 0$, and $|x| = -x$, if $x < 0$. Then the value of the integral

$$\int_{-2}^1 ([x] + 2)^{|x|} dx \text{ is}$$

- (A) $1 + \frac{1}{\log_e 2}$
 (B) $1 + \log_e \frac{1}{2}$
 (C) $2 - \log_e 2$
 (D) None of the above

Answer and Comments: (A). The absolute value function $|x|$ is continuous everywhere, but changes its formula at 0. The integral part function $[x]$ is discontinuous at all integers. Hence to evaluate the given integral, say I , we need to split the domain $[-2, 1]$ at -1 and 0 . (No need to split at the end points -2 and 1 .) So,

$$I = I_1 + I_2 + I_3 \quad (1)$$

where

$$I_1 = \int_{-2}^{-1} ([x] + 2)^{|x|} dx \quad (2)$$

$$I_2 = \int_{-1}^0 ([x] + 2)^{|x|} dx \quad (3)$$

$$\text{and } I_3 = \int_0^1 ([x] + 2)^{|x|} dx \quad (4)$$

We evaluate each of these three integrals, ignoring the discontinuities, if any, at the end-points since they do not affect the integral. On each of these subintervals, $[x]$ is constant with values -2 , -1 and 0 respectively. Also $|x| = -x$ for the first two subintervals and $|x| = x$ for the third. Hence

$$I_1 = \int_{-2}^{-1} 0^{-x} dx = 0 \quad (5)$$

$$I_2 = \int_{-1}^0 1^{-x} dx = \int_{-1}^0 1 dx = 1 \quad (6)$$

$$\text{and } I_3 = \int_0^1 2^x dx = \frac{1}{\ln 2} (2 - 1) = \frac{1}{\ln 2} \quad (7)$$

Adding, $I = 1 + \frac{1}{\ln 2} = 1 + \frac{1}{\log_2 2}$. ■

The main idea is to split the interval of integration into three parts. After that the calculations of the integrals are simple but repetitious and hence tedious, prone to numerical mistakes.

- Q.17 Let $f(x) = \frac{1}{1+|x-1|} + \frac{1}{1+|x+1|}$. Then the function f has
- (A) neither a local maximum nor a local minimum
 - (B) a local minimum at $x = 0$, but no local maximum
 - (C) local maxima at $x = \pm 1$, but no local minimum
 - (D) A local minimum at $x = 0$ and local maxima at $x = \pm 1$

Answer and Comments: (D). The function $|x|$ is an even function of x because $|-x| = |x|$. Hence its graph is symmetric about the y -axis. However, $|x-1|$ is not an even function of x . All we can say is that its graph is symmetric about the line $x = 1$. Same holds for $|x+1|$, where the line of symmetry is $x = -1$.

Interestingly, if we combine these two, we get an even function. Thus,

$$\begin{aligned} f(-x) &= \frac{1}{1+| -x-1|} + \frac{1}{1+| -x+1|} \\ &= \frac{1}{1+|x+1|} + \frac{1}{1+|x-1|} \\ &= f(x) \end{aligned} \tag{1}$$

Hence the graph of $y = f(x)$ is symmetric about the y -axis. This reduces by half the checking of many properties. For example, if there is a local maximum/minimum at some $\alpha \neq 0$, then at its reflection $-\alpha$ also there is a local maximum/minimum. What happens at 0 needs to be analysed separately.

So, first assume $x > 0$. The formula for $f(x)$ changes only at $x = 1$. So we have

$$f(x) = \begin{cases} \frac{1}{2-x} + \frac{1}{2+x} & \text{if } x < 1 \\ \frac{1}{x} + \frac{1}{x+2} & \text{if } x \geq 1 \end{cases} \tag{2}$$

For $x < 1$, the two terms add and give $f(x) = \frac{4}{4-x^2}$. As x increases, so does x^2 and $4-x^2$ decreases and further $\frac{4}{4-x^2}$ increases. So even without derivatives, we see that $f(x)$ is increasing on $[0, 1]$. However, for $x \in [1, \infty)$ both the summands are decreasing and hence $f(x)$ is

decreasing. Put together, the behaviour of $f(x)$ changes from increasing to decreasing at $x = 1$. Therefore it has a local maximum at 1. By symmetry about the y -axis, it has a local minimum at $x = -1$ too.

As for the behaviour of $f(x)$ at 0, we already know that it is increasing on $[0, 1)$. By symmetry, it is decreasing on $(-1, 0]$. Hence at 0, the behaviour changes from decreasing to increasing. So there is a local minimum at 0. All put together, there is a local minimum at 0 and local maxima at $x = \pm 1$. ■

A good problem on local maxima/minima. The answer would come quickly by looking at the graph. But it takes just about the same time to draw a good graph in the first place.

q.18 Let

$$f(x) = \begin{cases} -1, & \text{if } x < 0, \\ 0, & \text{if } x = 0, \\ 1, & \text{if } x > 0. \end{cases}$$

Then the function F defined by $F(x) = \int_{-5}^x f(t)dt$ is

- (A) not continuous
- (B) continuous, but nowhere differentiable
- (C) differentiable everywhere
- (D) differentiable everywhere except at 0

Answer and Comments: (D). $F(x)$ is defined by an integral. Such functions are always continuous. Further, they are differentiable wherever the integrand is continuous. So, in disguise, this is a question about testing the continuity of the function $f(x)$. Clearly, it is continuous everywhere except at 0. So $F(x)$ is differentiable everywhere except at 0. ■

The property quoted of functions defined by integrals is a consequence of the Fundamental Theorem of Calculus. It is good to see a question so directly based on it. For those who are not familiar with it, a direct answer to the present problem can be given by actually calculating $F(x)$. For $x \leq 0$, $F(x) = \int_{-5}^x -1dt = -(x+5)$. For $x > 0$, we must

split the interval of integration at 0. Doing so, $F(x) = \int_{-5}^0 -1 dt + \int_0^x 1 dt$ which equals $5 + x$. Hence

$$F(x) = \begin{cases} -x - 5, & \text{if } x \leq 0 \\ x + 5, & \text{if } 0 < x \end{cases} \quad (1)$$

As $F(x)$ is a linear function on $(-\infty, 0]$ as well as on $(0, \infty)$ it is differentiable at every $x \neq 0$. However, at 0 its left derivative is -1 while the right derivative is 1 . Hence it is not differentiable at 0. The same answer as above.

The given function $f(x)$ is often called the **signum function** and denoted by $\text{sgn}(x)$. It merely tells whether x is positive or negative. It can be used to express other functions which have different formulas depending upon whether the independent variable is positive or negative. For example, $|x|$ is simply $\text{sgn}(x)x$.

Q.19 For a real number x , let $f(x) = \int_{-20}^{20} g(t)g(x-t)dt$, where $g(x) = \begin{cases} 1, & \text{if } x \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$ Then $f(x)$ is equal to

(A) $\begin{cases} x, & \text{if } x \in [0, 1], \\ 2-x, & \text{if } x \in [1, 2], \\ 0, & \text{otherwise} \end{cases}$

(B) $\begin{cases} 1+x, & \text{if } x \in [0, 1], \\ 1-x, & \text{if } x \in [1, 2], \\ 0, & \text{otherwise} \end{cases}$

(C) $\begin{cases} 1, & \text{if } x \in (-20, 20), \\ 0, & \text{otherwise} \end{cases}$

(D) none of the above

Answer and Comments: (A). The last problem was about the derivative of a function defined by an integral. The present problem is to simply evaluate such a function. In the last problem, the answer could be obtained without actually computing the integrals if we use the FTC. In the present problem there is no choice.

Coming to the evaluation of $f(x)$, in the integral $\int_{-20}^{20} g(t)g(x-t)dt$, x is

fixed and for each such fixed x , we are integrating a function of t , viz. $g(t)g(x-t)$ over the interval $[-20, 20]$. The first factor vanishes for all $t \notin [0, 1]$ and equals 1 for all $t \in [0, 1]$. Splitting the interval $[-20, 20]$, into three parts, we get

$$\begin{aligned} f(x) &= \int_{-20}^0 g(t)g(x-t)dt + \int_0^1 g(t)g(x-t)dt + \int_1^{20} g(t)g(x-t)dt \\ &= \int_0^1 g(x-t)dt \end{aligned} \quad (1)$$

because the first and the third integrals vanish and in the second one, the first factor is identically 1.

To evaluate the integral that is now left, we put $u = x - t$. Then $dt = -du$. Also $u = x$ when $t = 0$ and $u = x - 1$ when $t = 1$. So,

$$f(x) = - \int_x^{x-1} g(u)du \quad (2)$$

As the upper limit of integration is smaller than the lower one, we reverse the order and get

$$f(x) = \int_{x-1}^x g(u)du \quad (3)$$

The value of $g(u)$ is either 0 or 1. Specifically, it equals 1 for that part of the interval $[x-1, x]$ which lies inside $[0, 1]$. Therefore, geometrically, $f(x)$ equals the length of the intersection of the two intervals $[x-1, x]$ and $[0, 1]$. (The intersection of two closed intervals is always an interval which may sometimes degenerate into a single point or even the empty set, depending upon the relative locations of the two intervals.) When $x < 0$ or $x - 1 > 1$ (i.e. $x > 2$) the entire interval $[x-1, x]$ is disjoint from $[0, 1]$ and hence $f(x) = 0$. But when x lies between 0 and 2, the length of the intersection increases linearly as x increases and reaches its maximum (when $x = 1$), but thereafter decreases linearly till $x = 2$ at which point its value is 0.

This verbal description can be expressed analytically as

$$f(x) = \int_{x-1}^x g(y)dy = \begin{cases} 0, & x \leq 0 \\ \int_0^x 1dy = x, & 0 < x < 1 \\ \int_{x-1}^1 1dy = 2 - x, & 1 \leq x < 2 \\ 0, & x \geq 2 \end{cases} \quad (4)$$

If we club together the first and the fourth formula, then this is exactly the same as (A). ■

An excellent problem testing various abilities. This is a rare problem where the part of integration *per se* is trivial because the integrand takes only two possible values. Art lies in breaking up the interval of integration suitably.

Q.20 Let

$$L = \lim_{n \rightarrow \infty} (n + 100)^{\frac{5}{\log_e(n-50)}}$$

Then

- (A) $2 \leq L \leq 16$
- (B) $16 \leq L \leq 32$
- (C) $32 \leq L \leq 243$
- (D) $L > 243$

Answer and Comments: (C). This is a sequence of powers where both the base and the exponent are functions of n . As $n \rightarrow \infty$, the base tends to ∞ but the exponent tends to 0. Therefore, the limit is an indeterminate type ∞^0 . The discrete variable can be replaced by a continuous variable x . The advantage is that certain concepts such as continuity, derivatives and certain theorems such as various versions of the l'Hôpital's rule can be used. (These theorems sometimes have some analogues for sequences. But they are not so well-known and easy to use.)

So, we let

$$M = \lim_{x \rightarrow \infty} (x + 100)^{\frac{5}{\ln(x-50)}} \quad (1)$$

and try to evaluate M . It may happen that M does not exist. In that case this approach fails. But in case it does, then L has to equal M . The first conversion we do is to take natural logarithms. Then

$$\begin{aligned} \ln M &= \lim_{x \rightarrow \infty} \frac{5}{\ln(x-50)} \ln(x+100) \\ &= 5 \lim_{x \rightarrow \infty} \frac{\ln(x+100)}{\ln(x-50)} \end{aligned} \quad (2)$$

It is intuitively obvious that for large x the numerator and the denominator are nearly equal and so the limit is 1. For a rigorous proof, we apply l'Hôpital's rule and take the ratio of their derivatives (which make sense now that x is a continuous variable). So,

$$\begin{aligned}\ln M &= 5 \lim_{x \rightarrow \infty} \frac{x - 50}{x + 100} \\ &= 5 \times 1 = 5\end{aligned}\tag{3}$$

Hence

$$L = e^M = e^5\tag{4}$$

We now have to see which of the given intervals contain e^5 . The approximate value of e is 2.71828. So, definitely $2 < e < 3$. Therefore $2^5 < e^5 < 3^5$, i.e. $32 < L < 243$. ■

A routine problem once it is realised which method to adopt.

- Q.21 Let a, b, c, d be positive integers such that the product $abcd = 999$. Then the number of different ordered 4-tuples (a, b, c, d) is
- (A) 20
 - (B) 48
 - (C) 80
 - (D) 84

Answer and Comments: (C). The prime power factorisation of 999 is $3^3 \times 37^1$. Hence the factors of 999 are of the form $3^r 37^s$, for $0 \leq r \leq 3$ and $0 \leq s \leq 1$. So assume $a = 3^{r_1} 37^{s_1}$, $b = 3^{r_2} 37^{s_2}$, $c = 3^{r_3} 37^{s_3}$ and $d = 3^{r_4} 37^{s_4}$ where $r_1, r_2, r_3, r_4 \in \{0, 1, 2, 3\}$ and $s_1, s_2, s_3, s_4 \in \{0, 1\}$. The requirement that $abcd = 999$ is equivalent to

$$r_1 + r_2 + r_3 + r_4 = 3\tag{1}$$

$$\text{and } s_1 + s_2 + s_3 + s_4 = 1\tag{2}$$

Any solution of (1), clubbed with any solution of (2) give a factorisation of 999 in the form $abcd$ and this correspondence is a bijection. Therefore, the problem is now reduced to counting the numbers of solutions of (1) and (2) each and multiplying the two counts.

It is easier to solve (2). Exactly one s_i is 1 and the others 0 each. This can happen in $\binom{4}{1} = 4$ ways. There are several methods to count the number of solutions to (1). The most elementary (given in Comment No. 5 of Chapter 1) is to recognise every solution (r_1, r_2, r_3, r_4) as an arrangement of 3 identical balls and 3 identical walls in a row so that r_1, r_2, r_3, r_4 are the numbers of balls in the four boxes formed by the walls. There are $\binom{6}{3} = 20$ such solutions. Hence the total number of solutions to (1) and (2) together is $20 \times 4 = 80$. This is the number of different ordered 4-tuples (a, b, c, d) such that $abcd = 999$. ■

A standard but good problem on counting. Number theory involved is minimal. Prime power factorisation was also needed in Q.9 above.

- Q.22 In a certain test there are n questions. At least i questions were wrongly answered by 2^{n-i} students, where $i = 1, 2, \dots, n$. If the total number of wrong answers given by all students is 2047, then n is equal to
- (A) 10
 - (B) 11
 - (C) 12
 - (D) 13

Answer and Comments: (B). We are *not* given how many students wrongly answer exactly i questions. But we can find out. The idea is similar to that of measuring how high a person can jump. You set the bar at some initial height, say 30 inches above the ground. Let x be the number of persons who clear it. Some of these persons may be able to jump higher than 30 inches too. So you raise the bar by one inch (or some such agreed unit). Let y be the number of persons who can clear it. Obviously $y \leq x$. Some of these y persons may be able to jump even higher. But definitely $x - y$ is the number persons who can jump 30 inches but no higher (ignoring increases of heights less than an inch).

The same logic, which is essentially common sense, is useful here. $2^{n-i} - 2^{n-i-1}$ is the number of students who answer exactly i questions wrongly. This is valid for $i = 1, 2, \dots, n-1$. However, for $i = n$, the data says that 1 student answered n questions wrongly. But since n is the total number of questions, there cannot be any student who

answered $n + 1$ questions wrongly, i.e. the number of such students is 0 and not 2^{n-n-1} (which is actually a fraction).

From this, the total number of wrong answers, say w , is the sum

$$w = \sum_{i=1}^{n-1} (2^{n-i} - 2^{n-i-1})i + n \quad (1)$$

$$= \sum_{i=1}^{n-1} (2^{n-i-1})i + n \quad (2)$$

(The last term n is added because of the exclusion of $i = n$ from the summation and the lone student who answers all n questions wrongly.)

There are binomial identities which give each of the sums. We can apply one such identity to evaluate the sum in (2). But there is a better way. If we simply write out the sum in (1) in full, it becomes

$$w = 1(2^{n-1} - 2^{n-2}) + 2(2^{n-2} - 2^{n-3}) + 3(2^{n-3} - 2^{n-4}) + \dots \\ + (n-1)(2^1 - 2^0) + n \quad (3)$$

This is almost like a telescopic series, except that the second term in each group has a coefficient which is 1 less than the coefficient of the first term of the next group. Therefore we are left with

$$w = 2^{n-1} + 2^{n-2} + \dots + 2 + 1 \quad (4)$$

$$= 2^n - 1 \quad (5)$$

as the series in (4) is a geometric series.

But w is given to be 2047. That gives $2^n = 2048$. Since $2048 = 2^{11}$ we must have $n = 11$. ■

An interesting problem where the use of binomial identities is possible but not mandatory. The real crux of the problem is to translate the data which gives an upper bound on the number of students who answer a certain number of questions wrongly to a formula for the exact number of students who answer a certain number of questions wrongly. And the logic is mote common sense than mathematics. Other analogies are possible to illustrate the thought involved. Suppose, for example, that the population is divided into income slabs in increasing

order. If we know that 45% of the population is in or above some slab but only 37% of it is in or above the next slab, then we know that exactly 8% of the population is in that particular slab.

- Q.23 Let $n \geq 3$. There are n straight lines in a plane, no two of which are parallel and no three pass through a common point. Their points of intersection are joined. Then the number of fresh line segments thus created is

- (A) $\frac{n(n-1)(n-2)}{8}$
- (B) $\frac{n(n-1)(n-2)(n-3)}{6}$
- (C) $\frac{n(n-1)(n-2)(n-3)}{8}$
- (D) None of the above

Answer and Comments: (C). With three lines, say, L_1, L_2 and L_3 , we get a triangle (as no two are parallel and all three are not concurrent.) So, there are 3 lines and 3 segments. Number the remaining lines as $L_4, L_5, \dots, L_n$ in some arbitrary but fixed manner. Since every two of these lines intersect, the total number of points of intersection is $\binom{n}{2} = \frac{n(n-1)}{2}$. We get a segment by joining any two (distinct) points of intersection. However, some of these segments were already there. On each of the n lines, there are $n-1$ points of its intersection and any segment obtained by joining any two of them was already there. (Here it may happen that some segments are overlapping. Indeed some segments may be contained in some others. Still they are to be regarded as distinct.) So each of the n lines contains $\binom{n-1}{2}$ 'old' segments.

As a result, the problem boils down to counting the number of all pairs of the $n(n-1)/2$ of intersections and subtracting $n \times \binom{n-1}{2}$. This is the number, say a_n , of the number of fresh line segments. That is,

$$a_n = b_n - c_n \tag{1}$$

where

$$b_n = \binom{n(n-1)/2}{2} \text{ and } c_n = n \times \binom{n-1}{2} \tag{2}$$

We simplify each of these two.

$$\begin{aligned} b_n &= \frac{\frac{n(n-1)}{2} \left(\frac{n(n-1)}{2} - 1 \right)}{2} \\ &= \frac{n(n-1)(n-2)(n+1)}{8} \end{aligned} \quad (3)$$

where in the last step we have used $n(n-1) - 2 = n^2 - n - 2 = (n-2)(n+1)$.

It is much easier to simplify c_n .

$$c_n = \frac{n(n-1)(n-2)}{2} \quad (4)$$

Hence

$$\begin{aligned} a_n &= \frac{n(n-1)(n-2)(n+1)}{8} - \frac{n(n-1)(n-2)}{2} \\ &= \frac{n(n-1)(n-2)}{8} - [(n+1) - 4] \\ &= \frac{n(n-1)(n-2)(n-3)}{8} \end{aligned} \quad (5)$$

which tallies with (C). ■

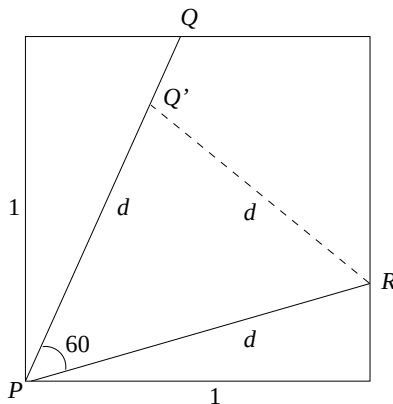
More a test of simplification of complicated algebraic expression. The falsity of the options (A) and (B) could be proved by working out special cases such as $n = 4$. If (D) were also of this type, a candidate could answer correctly by eliminating all other options. Probably to preclude such a sneaky answer, the paper setters have given the last option as ‘None of the above’.

Q.24 Let d be the side length of the largest possible triangle that can be put inside a square of side length 1. Then

$$(A) \ d < 1 \quad (B) \ d = 1 \quad (C) \ 1 < d < \frac{2}{3^{1/4}} \quad (D) \ d \geq \frac{2}{3^{1/4}}$$

Answer and Comments: (C). It would be great if we can calculate d exactly. For then we only need to compare it directly with the given numbers. But calculating d exactly is not likely to be easy as we shall

elaborate soon. But the problem does *not* ask the exact value of d . That suggests that there may be a short cut. We can easily draw equilateral triangles with one vertex, say P at a corner and the other two, say Q and R , on the sides opposite to the corner, in such a way that $\angle QPR = 60^\circ$. If $PQ = PR$, we already have an equilateral triangle. If $PQ > PR$ (say), then on PQ mark Q' so that $PQ' = PR$. Then $\triangle PQ'R$ is equilateral. The side is bigger than 1, the side of the square.



This already rules out options (A) and (B). Also the first half of (C) is trivially true. Therefore the problem now reduces to comparing d with $\frac{2}{3^{1/4}}$, or equivalently, to comparing d^2 with $\frac{4}{\sqrt{3}}$. This is more inviting since d^2 arises naturally in the expression for the area of an equilateral triangle with side d . Specifically, the area is $\frac{\sqrt{3}d^2}{4}$ as can be seen in various ways, the easiest probably being that two sides of each of length d include an angle 60° . But this number, being the area of a subset of the unit square, is less than 1. Therefore $\sqrt{3}d^2 < 4$. Taking square roots, $d < \frac{2}{3^{1/4}}$. Hence (C) is true and (D) false. ■

The question is more of a test of the ability to spot possible short cuts keeping in mind exactly what is asked. The exact value of d is *not* needed to answer the question as it is. It comes out as $\sqrt{6} - \sqrt{2}$. If we can get hold of it, we can directly compare $\sqrt{6} - \sqrt{2}$ with $\frac{2}{3^{1/4}}$. Squaring both the sides the comparison reduces to that between $8 - 4\sqrt{3}$ and $\frac{4}{\sqrt{3}}$, i.e. between $2 - \sqrt{3}$ and $\frac{1}{\sqrt{3}}$ and hence finally to that between $2\sqrt{3} - 3$ and 1. With some more reduction, we have to compare $\sqrt{3}$ with 2. Clearly, the L.H.S. is smaller. Therefore d is less than $\frac{2}{3^{1/4}}$.

However, calculating d solely for this purpose is like using a sledge hammer to kill a fly. Nevertheless, for the sake of completeness, we include a proof of this in the Appendix at the end of this commentary. Predictably, the proof is quite involved. In general proofs of geometric optimisations are so. Very rarely, a short proof can be given using some other known optimisation result. For example, using that the shortest path between two points is a straight line, one can, for a given triangle ABC , find a point for which $PA+PB+PC$ is minimum. (See Exercise (14.16).) It would be interesting to have such a proof for the present problem.

Q.25 Let

$$a_n = \frac{n^2}{\sqrt{n^6+1}} + \frac{n^2}{\sqrt{n^6+2}} + \dots + \frac{n^2}{\sqrt{n^6+2}}, \quad n \geq 1.$$

Then, $\lim_{n \rightarrow \infty} a_n$

(A) does not exist (B) equals 1 (C) equals e (D) equals $\frac{1}{e}$

Answer and Comments: (B). a_n is the sum of n terms which are in descending order. So a_n lies between n times the first term and n times the last term. That is,

$$\frac{n^3}{\sqrt{n^6+n}} \leq a_n \leq \frac{n^3}{\sqrt{n^6+1}} \quad (1)$$

We claim that as $n \rightarrow \infty$, both the first and the last expression tend to 1. This is very obvious for the last term because in it the denominator is only slightly bigger than the numerator. However, for a rigorous proof,

$$\lim_{n \rightarrow \infty} \frac{n^3}{\sqrt{n^6+1}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+1/n^6}} \quad (2)$$

$$\begin{aligned} &= \frac{1}{\sqrt{1+\lim_{n \rightarrow \infty}(1/n^6)}} \\ &= \frac{1}{\sqrt{1+0}} = 1 \end{aligned} \quad (3)$$

where (2) was obtained by dividing both the numerator and the denominator of the L.H.S. by n^3 .

For limit of the first term in (1), the calculations are only marginally different. Even though n is bigger than 1, as compared to n^6 it is insignificant.

$$\lim_{n \rightarrow \infty} \frac{n^3}{\sqrt{n^6 + n}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1 + 1/n^5}} \quad (4)$$

$$\begin{aligned} &= \frac{1}{\sqrt{1 + \lim_{n \rightarrow \infty} (1/n^5)}} \\ &= \frac{1}{\sqrt{1 + 0}} = 1 \end{aligned} \quad (5)$$

The result follows by applying the Sandwich theorem (also called the Squeeze theorem sometimes) to (1). ■

Q.26 Let n be a positive integer. The value of $\sum_{k=0}^n \tan^{-1} \frac{1}{k^2 + k + 1}$ is

(A) $\tan^{-1}(n+1)$ (B) $\tan^{-1}\left(\frac{1}{n+1}\right)$ (C) $\tan^{-1} n$ (D) $\tan^{-1}\left(\frac{1}{n}\right)$

Answer and Comments: (A). The well-known formula

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \quad (1)$$

translates into

$$\tan^{-1} \left(\frac{x - y}{1 + xy} \right) = \tan^{-1} x - \tan^{-1} y \quad (2)$$

if we put $x = \tan \alpha$ and $y = \tan \beta$. This is very handy for converting the arctan of a ratio in terms of arctans of simpler terms. The ratio, of course, has to be expressible in a certain form.

The present problem deals with a sum of the form $\sum_{k=0}^n T_k$ where $T_k = \tan^{-1} \frac{1}{k^2 + k + 1}$. If we write $\frac{1}{k^2 + k + 1}$ as $\frac{(k+1)-k}{1+k(k+1)}$ and apply (2), then T_k

becomes $\tan^{-1}(k+1) - \tan^{-1} k$. Hence the given sum, say, S is

$$S = \sum_{k=0}^n \tan^{-1}(k+1) - \tan^{-1} k \quad (3)$$

This is a telescopic series with sum $\tan^{-1}(n+1) - \tan^{-1} 0$ i.e. $\tan^{-1}(n+1)$. ■

Problems based on this trick have been asked several times in JEE.

Q.27 Let $f(x) = (x^2 + 18)(x - 4)x(x + 4) - 2$. Then

- (A) f has exactly one real root
- (B) f has exactly 3 distinct real roots
- (C) f has 5 distinct real roots
- (D) f has a repeated root

Answer and Comments: (B). Being an odd degree real polynomial, $f(x)$ has at least one real root. To see if it has any more, we first write $f(x)$ as

$$\begin{aligned} f(x) &= x(x^2 + 18)(x^2 - 16) - 2 \\ &= x^5 + 2x^3 - 288x - 2 \end{aligned} \quad (1)$$

If there are any other (real) roots, by Rolle's theorem, the derivative must vanish at least once. In the present case, this means that the equation

$$f'(x) = 5x^4 + 6x^2 - 288 = 0 \quad (2)$$

has at least one solution. Although this is a quartic, i.e. a fourth degree equation, there are no odd degree terms. Therefore if we put $u = x^2$, it becomes a quadratic in u .

$$5u^2 + 6u - 288 = 0 \quad (3)$$

This has two real roots. As their product is negative, only one of them, say α is positive. The two numbers $\pm\sqrt{\alpha}$ are the only two real roots of (2). Therefore, $f(x)$ has exactly three real roots, one in each of the intervals $(-\infty, -\sqrt{\alpha})$, $(-\sqrt{\alpha}, \sqrt{\alpha})$ and $(\sqrt{\alpha}, \infty)$. ■

The question is conceptually similar to but computationally a little more involved than Q.7.

Q.28 Let k be a positive integer and $f(x) = e^x - 1$. Then

$$\lim_{x \rightarrow 0} \frac{f(x) + f\left(\frac{x}{2}\right) + f\left(\frac{x}{2^2}\right) + \dots + f\left(\frac{x}{2^k}\right)}{x}$$

is equal to

(A) $2 - \frac{1}{2^k}$ (B) $2 - \frac{1}{2^{k+1}}$ (C) k (D) $2^{k+1} - 1$

Answer and Comments: (A). The expression whose limit is taken is a sum of $k + 1$ functions where k is a fixed positive integer. So, we might as well take the limit of each term and add. For each of these $k + 1$ limits, simply the definition of the derivative at $x = 0$ would do.

For $r = 0, 1, 2, \dots, k$, we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{x/2^r} - 1}{x} &= \left(\frac{d}{dx} (e^{x/2^r}) \right)_{x=0} \\ &= \frac{1}{2^r} \end{aligned} \quad (1)$$

Hence the given limit is the sum $\sum_{r=0}^k \frac{1}{2^r}$. As the terms are in G.P. with common ratio $\frac{1}{2}$, this equals $\frac{1 - \frac{1}{2^{k+1}}}{1 - \frac{1}{2}}$ and hence $2 - \frac{1}{2^k}$. ■

A simple problem, combining exponential function with geometric series.

Q.29 Let

$$a_n = \frac{2^3 - 1}{2^3 + 1} \times \frac{3^3 - 1}{3^3 + 1} \times \dots \times \frac{n^3 - 1}{n^3 + 1}, \quad n \geq 2$$

. Then $\lim_{n \rightarrow \infty} a_n$

(A) does not exist (B) equals $\frac{2}{3}$ (C) equals 1 (D) equals $\frac{1}{2}$

Answer and Comments: (B). In a compact form,

$$a_n = \prod_{r=2}^n \frac{r^3 - 1}{r^3 + 1} \quad (1)$$

Products of terms of a sequence appear much more rarely than their sums. So unlike formulas for sums, there are hardly any formulas to give a handy expression for a product. We have to work fresh to express a_n in a more manageable form. The numerator and the denominator of the r -th factor factors as $(r-1)(r^2+r+1)$ and $(r+1)(r^2-r+1)$ respectively. So we can write

$$a_n = b_n c_n \quad (2)$$

where

$$b_n = \prod_{r=2}^n \frac{r-1}{r+1} \quad (3)$$

$$\text{and } c_n = \prod_{r=2}^n \frac{r^2+r+1}{r^2-r+1} \quad (4)$$

Unlike a_n , b_n has an obvious simplification. The factors in the numerator, except for the first factor 2, all appear in the denominator too and hence get canceled with all except the last two factors of the denominator. Hence

$$b_n = \frac{2}{n(n+1)} \quad (5)$$

However, in c_n similar cancellations are not so obvious. To make them so, we note that the r -th factor of the numerator, viz. r^2+r+1 , can be rewritten as $(r+1)^2 - (r+1) + 1$. This is precisely the $(r+1)$ -th factor of the denominator. Canceling them, we are left with only the last factor of the numerator and the first factor of the denominator. Hence

$$c_n = \frac{n^2+n+1}{3} \quad (6)$$

From (4), (5) and (6),

$$a_n = \frac{2}{3} \frac{n^2+n+1}{n(n+1)} \quad (7)$$

As $n \rightarrow \infty$, the second factor tends to 1, being the ratio of two quadratic polynomials in n with the same leading coefficient. Hence $a_n \rightarrow \frac{2}{3}$ as $n \rightarrow \infty$. ■

A rather tricky problem, more on algebra than on calculus.

- Q.30 A subset $\{u_1, u_2, u_3, u_4, u_5\}$ of the first 90 positive integers can be selected in $\binom{90}{5}$ ways. Let $u_{\max} = \max\{u_1, u_2, u_3, u_4, u_5\}$ and $u_{\min} = \min\{u_1, u_2, u_3, u_4, u_5\}$. Then the arithmetic mean of $u_{\max} + u_{\min}$ over all such subsets is
- (A) 45
 - (B) 46
 - (C) 89
 - (D) 91

Answer and Comments: (D). We are given a set, say T , of all 5-element subsets of $\{1, 2, \dots, 90\}$ having five elements. We have two integer valued functions $u_{\max}$ and $u_{\min}$ defined on T and we are asked the average value of their sum.

When we want the average value or the mean of a function, say f , the standard practice is to sum the values of f at each point of the set and divide the sum by the cardinality of, i.e. the number of elements in, the set. But sometimes a slightly different method is adopted. Instead of summing the values of the function at all points of the domain, the domain is divided into classes on each of which the function has the same value, say, v . Let f_v be the number of elements of the domain at which the function has value v . This number is called the **frequency** of the value v . By classifying elements according to the values of the function, in effect, we are preparing the **frequency distribution** of the function. Then the average of the function is simply the ratio $\frac{\sum_v f_v v}{\sum_v f_v}$ where both the sums range over the set of possible values of the function.

Given any typical element, say $u = \{u_1, u_2, u_3, u_4, u_5\}$ of T , (where we assume, without loss of generality that the elements are in ascending order) clearly its maximum can be any integer from 5 to 90 and its minimum can be any integer from 1 to 86. For a given r from 1 to 86, the number of 5-element subsets of $\{1, 2, \dots, 90\}$ which have r as their minimum is $\binom{90-r}{4}$ because the remaining four elements must come from the set $\{r+1, r+2, \dots, 90\}$. Similarly, the number of subsets having r as their maximum is $\binom{r-1}{4}$ because the remaining four

elements must come from $\{1, 2, \dots, r-1\}$. Therefore if we denote the average minimum and the average maximum by m and M respectively, then we have

$$m = \frac{S_1}{|T|} \tag{1}$$

$$\text{and } M = \frac{S_2}{|T|} \tag{2}$$

where $|T|$ is the total number of 5-element subsets of $\{1, 2, \dots, 90\}$, i.e. $|T| = \binom{90}{5}$ and S_1, S_2 are, respectively the sums

$$S_1 = \sum_{r=1}^{90} r \binom{90-r}{4} \quad \text{and} \quad S_2 = \sum_{r=1}^{90} r \binom{r-1}{4}$$

(There is no harm in letting r vary from 1 to 90 in both the sums because the binomial coefficient $\binom{m}{n}$ is 0 if $m < n$.)

There is a variety of methods to evaluate these sums S_1 and S_2 . (See the first seven Comments in Chapter 5.) Putting these into (1) and (2) and adding the two will give the answer to the problem.

But there is a far more elegant way. A perceptive candidate will notice that we are *not* interested in the values of m and M *per se* but only in their sum $m + M$. Recall that m is the sum of the minima of all 5-element subsets of $\{1, 2, \dots, 90\}$ while M is the sum of their maxima. So far we summed all minima and all maxima separately. But what if we, for each 5-element subset of $\{1, 2, \dots, 90\}$, add its minimum and its maximum and then add these sums as the subset ranges over T ? Then we would get the same answer. As a real life analogy, suppose we want to find the average individual income of a community with 100 married couples. We can either add all husbands' incomes, then all wives' incomes and add the two sums. Or, for each couple, we can take its total income and add all these 100 sums. Either way, we get the average income by dividing the grand total by 200.

The trouble is that there is no handy way to express the sum of the minimum and maximum of a 5-element subset of $\{1, 2, \dots, 90\}$. For example, if the subset is $\{2, 5, 49, 74, 82\}$ then the minimum is 2, the maximum is 82 and their sum is 84. For the subset $\{3, 9, 17, 44, 65\}$, it is 68. And it will vary a lot. There is no formula for it.

But what if we add the minimum of a subset and the maximum of some other subset related to it? As long as this relationship is bijective, we would get the same sum $m + M$ by adding the incomes of all these ‘odd couples’. The success of this approach depends upon choosing this relationship cleverly.

We mention one such relationship. For any $x \in \{1, 2, \dots, 90\}$, $91 - x$ is also an element of it and the two always add to 91. Geometrically, they are reflections of each other w.r.t. the mid-point 45.5. So, let us define a function $r : \{1, 2, \dots, 90\} \longrightarrow \{1, 2, \dots, 90\}$ by $r(x) = 91 - x$. We call this the reflection. It is clearly a bijection. In fact, it is its own inverse. It is order reversing, that is, if $x < y$, then $r(y) < r(x)$. As a result, for any subset S of $\{1, 2, \dots, 90\}$, the maximum of its reflection is the reflection of the minimum of S . It follows that no matter which subset you take, the sum of its minimum and the maximum of its reflection is a constant. As an example, take the 5-element subset $\{2, 5, 49, 74, 82\}$ considered above. Its reflection is $\{89, 86, 42, 17, 9\}$. The minimum of the former is 2 while the maximum of the reflection is 89. Their sum is 91.

So, if we pair every 5-element subset of $\{1, 2, \dots, 90\}$ with its reflection, the sum of the minimum of one and the maximum of the other is a constant 91. Doing this for every 5-element subset of $\{1, 2, \dots, 90\}$, we now see effortlessly that $m + M$ is simply $91 \times |T|$. Dividing this by $|T|$, we get that $u_{\max} + u_{\min}$ is 91. ■

The reflection principle appears in other connections too. The well-known formula for integrals

$$\int_0^a f(x)dx = \int_0^a f(a - x)dx \quad (3)$$

is an instance of it. Here the reflection is the function $r(x) = a - x$. (The great mathematician Gauss, when a school boy with age 7, found the sum $1 + 2 + 3 + \dots + 100$ instantaneously, by clubbing it with its reflection $100 + 99 + \dots + 2 + 1$. Q.16 of Paper 2 of JEE Advanced 2020 was also based on the reflection principle.)

The paper-setters deserve to be commended for asking a question which can be done elegantly using the reflection principle. But, once again, asking it as an MCQ is killing its beauty.

ISI BStat-BMath-UGB-2025 Paper 2 with Comments

Each question carries **TEN** points.

- Q. 1 Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable and $|f'(x)| < \frac{1}{2}$ for all $x \in \mathbb{R}$. Show that for some $x_0 \in \mathbb{R}$, $f(x_0) = x_0$.

Solution and Comments: If the assertion is false, then because of continuity of $f(x)$ we must have either $f(x) > x$ for all $x \in \mathbb{R}$ or $f(x) < x$ for all $x \in \mathbb{R}$. Without loss of generality we assume the first possibility, as otherwise we replace $f(x)$ by $-f(x)$.

Fix any $a_0 \in \mathbb{R}$. Define a sequence $\{a_n\}_{n \geq 1}$ recursively by

$$a_n = f(a_{n-1}) \quad (1)$$

for $n \geq 1$. Since $a_1 = f(a_0) > a_0$, and $a_2 = f(a_1) > a_1$ and, more generally, $a_{n+1} = f(a_n) > a_n$ for all n , we get that the sequence $\{a_n\}$ is monotonically increasing.

We claim that this sequence is convergent. For this it suffices to show that it is bounded above. We apply the Lagrange's MVT to the interval $[a_{n-1}, a_n]$ to get

$$a_{n+1} - a_n = f(a_n) - f(a_{n-1}) = f'(x_n)(a_n - a_{n-1}) \quad (2)$$

for some $x_n \in (a_{n-1}, a_n)$. Since $a_{n+1} - a_n$ and $a_n - a_{n-1}$ are both positive, so is $f'(x_n)$. Further we are given that its absolute value is less than $\frac{1}{2}$. Therefore,

$$a_{n+1} - a_n < \frac{1}{2}(a_n - a_{n-1}) \quad (3)$$

Verbally, even though the sequence $\{a_n\}_{n \geq 0}$ is monotonically increasing, the difference between any two consecutive terms is less than half of the difference between the earlier terms. Using (3) repeatedly and induction on n , we get that for every $n \geq 1$,

$$a_n - a_{n-1} \leq \left(\frac{1}{2}\right)^{n-1}(a_1 - a_0) \quad (4)$$

(with equality holding only for $n = 1$.) Hence

$$a_n - a_0 = \sum_{k=0}^{n-1} (a_{k+1} - a_k) \leq \sum_{k=0}^{n-1} \left(\frac{1}{2}\right)^k (a_1 - a_0) \quad (5)$$

Since the geometric series $\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k$ is convergent, the sequence $\{a_n\}$ is bounded above. Therefore it is convergent. Denote its limit by L . Then by continuity of $f(x)$ $f(a_n) \rightarrow f(L)$ as $n \rightarrow \infty$. But $f(a_n) = a_{n+1}$ by construction. Now as n tends to *infinity*, so does $n + 1$. Hence $\lim_{n \rightarrow \infty} a_{n+1}$ also equals L . Hence $f(L) = L$, a contradiction. This contradiction arose because of our assumption that there does not exist any $x_0 \in \mathbb{R}$ for which $f(x_0) = x_0$. That proves the assertion. ■

A point x_0 for which $f(x_0) = x_0$ is called, quite appropriately, a **fixed point** of the function f . Thus we have proved that a function with the given properties has at least one fixed point. Actually, we have proved something more. Starting with any $a_0 \in \mathbb{R}$, we have shown that the sequence obtained by applying iterates of f to a_0 converges to a fixed point of f .

There is nothing special about $\frac{1}{2}$. If, instead, we have some real number $\alpha < 1$ such that $|f'(x)| < \alpha$ for all $x \in \mathbb{R}$, then also the argument above will go through because the geometric series $\sum_{n=0}^{\infty} \alpha^n$ is convergent. In fact, we do not even need differentiability of f . If f satisfies the condition that there exists some $\alpha < 1$ such that for all $x, y \in \mathbb{R}$,

$$|f(x) - f(y)| \leq \alpha |x - y| \quad (6)$$

then also the argument above shows that f has a fixed point. Of course, we must first ensure continuity of f . But that follows easily from (6). Suggestively, such a function is called a **contraction** because it contracts the objects in its domain by reducing the distances between points. The result we have proved is therefore called the **contraction fixed point theorem**. (An example of a continuous, non-differentiable contraction is the function $f(x) = \alpha|x|$ for $0 \leq \alpha < 1$.)

Note that we cannot allow α to equal 1. A simple counterexample is the function $f(x) = x + 1$. It satisfies (6) with $\alpha = 1$. But has no fixed points.

Q. 2 If the interior angles of a triangle ABC satisfy the equality

$$\sin^2 A + \sin^2 B + \sin^2 C = 2(\cos^2 A + \cos^2 B + \cos^2 C),$$

prove that the triangle must have a right angle.

Solution and Comments: There are numerous identities about the sums and products of the trigonometric functions of the angles A, B, C of a triangle ABC . Many of them only require that $\angle A + \angle B + \angle C = \pi$ and hence have a wider applicability. But generally they are used only for triangles. In the present problem, we apply one such identity, viz.

$$\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2 \cos A \cos B \cos C \quad (1)$$

Before applying this, let us recast the given equality slightly by writing $\cos^2 A$ as $1 - \sin^2 A$ and similarly for the other two terms. Then our data becomes $3(\sin^2 A + \sin^2 B + \sin^2 C) = 6$, i.e.

$$\sin^2 A + \sin^2 B + \sin^2 C = 2 \quad (2)$$

Application of (1) gives

$$\cos A \cos B \cos C = 0 \quad (3)$$

Hence at least one factor vanishes and the corresponding angle is a right angle. ■

A very simple problem once you know the right trick to apply. Instead of allotting all ten marks to prove the result, it would have been better to split the question into two parts, the first asking for a proof of this identity (which is not so trivial) and the second part its application (which is really trivial).

Q. 3 Suppose $f : [0, 1] \rightarrow \mathbb{R}$ is differentiable with $f(0) = 0$. If $|f'(x)| \leq f(x)$ for all $x \in [0, 1]$, then show that $f(x) = 0$ for all x .

Solution and Comments: The given inequality implies, in particular, that $f'(x) \leq f(x)$ for all $x \in [0, 1]$. We can rewrite this as

$$f'(x) - f(x) \leq 0 \quad (1)$$

for all $x \in [0, 1]$. We multiply the L.H.S. by e^{-x} which is positive for all real x . Hence

$$e^{-x}(f'(x) - f(x)) \leq 0 \quad (2)$$

for all $x \in [0, 1]$. Now comes the key observation that the L.H.S. is nothing but the derivative of the function $e^{-x}f(x)$. Call this product function as $g(x)$. Then the derivative of g is less than or equal to 0 for all $x \in [0, 1]$. So by Lagrange's MVT, $g(x)$ is monotonically decreasing on $[0, 1]$. Since $g(0) = e^{-0}f(0) = 0$, we get $g(x) \leq g(0) = 0$ for all $x \in [0, 1]$. So $e^{-x}f(x) \leq 0$ for all $x \in [0, 1]$. But the factor e^{-x} is always positive. So the only conclusion is that $f(x) \leq 0$ for all $x \in [0, 1]$. But on the other hand, since the absolute value is always non-negative, the hypothesis $|f'(x)| \leq f(x)$ for all $x \in [0, 1]$ implies that $f(x) \geq 0$ for all $x \in [0, 1]$. Both these statements can hold together only if $f(x)$ is identically 0 on $[0, 1]$. ■

A tricky problem where the trick is to multiply inequality (1) by e^{-x} and recognise the L.H.S. as the derivative of some function.

The problem is strikingly similar to the JEE 2001 given in Comment No. 13 of Chapter 19. Those who have seen that or some similar problem will have an easier time.

- Q. 4 Let $S^1 = \{z \in \mathbb{C}, |z| = 1\}$ be the unit circle in the complex plane. Let $f : S^1 \rightarrow S^1$ be the map given by $f(z) = z^2$. We define $f^{(1)} := f$ and $f^{(k+1)} = f \circ f^{(k)}$ for $k \geq 1$. The smallest integer n such that $f^{(n)}(z) = z$ is called the *period* of z . Determine the total number of points in S^1 of period 2025. (Hint: $2025 = 3^4 \times 5^2$.)

Solution and Comments: In grammar, a period is a punctuation mark that signifies the end of a sentence. Hence figuratively the word is sometimes used to indicate a resolute dead stop to something. Thus, "I am not coming. Period." means "My decision is final and no more discussion on it."

But in mathematics and more generally, in sciences, regular repetition is the essential idea of a period. Familiar examples are the periods of trigonometric functions or the period of the fractional part function. (In these examples, we talk of the period of a function. But in the problem we talk of the period of a given point in its domain. To go by this difference, the sine function has a period π at every multiple of π but the period 2π at all other points.)

In residue arithmetic, the period of a residue class is the number of times some action (such as taking the higher power or the next integral multiple) has to be carried on it to get back to the original value and thereafter keep getting the same values in a cycle. In biology there is the menstrual period or the mating periods of various species. In astronomy, there are periodic motions of celestial bodies.

The concept of a period defined in this problem is not a common one. Also it is likely to be confused with another related, and more commonly used concept defined for elements of S^1 , (i.e. for complex numbers of unit magnitude) by taking successive powers. For example, if $z = i$, then successive powers recur in a cycle of length 4. So, 4 is also a sort of period of i . But it is called the **order** of i . In general, for every positive integer k , if we let $\alpha = e^{2\pi i/k}$, then α has order k and k is the smallest positive integer such that the k -th power of α equals 1. If β is some power of α , say α^r , then also $\beta^k = \alpha^{rk} = (\alpha^k)^r = 1$. But the order of β may be less than k . For example, for $k = 4$, $\alpha = i$ and the order of $i^2 = -1$ is only 2 and not 4. In general all we can say is that the order of α^r divides k . (The exact order of β is k/d , where d is the g.c.d. of r and k .)

In this problem too, the period of a complex number z is defined in terms of when a certain sequence of powers of z encounters z and thereafter recurs. But the exponents in these powers are not in an A.P. as in the case of the definition of the order, but in a G.P. of common ratio 2. By induction on n it follows easily that for every positive integer n ,

$$f^{(n)}(z) = z^{2^n} \tag{1}$$

To prove this we merely note that $f^{(n+1)}(z) = f(f^{(n)}(z)) = (f^{(n)}(z))^2$ and $(z^{2^n})^2 = z^{2^n \times 2} = z^{2^{n+1}}$.

So, the period of z is the smallest positive integer n for which z^{2^n} equals z . As this is an unfamiliar concept, before tackling the problem, let us calculate the periods of some complex numbers. Clearly the period of 1 is 1. In fact, 1 is the only non-zero complex number which satisfies $z^2 = z$. Note, however, that the complex number -1 has no period because $(-1)^2 = 1, (-1)^4 = 1, (-1)^8 = 1$ and indeed all even powers of -1 equal 1. The complex number i has no period either, for $i^2 = -1, i^4 = 1, i^8 = 1$ and so are all powers with exponents of the form 2^n for $n > 1$. The smallest power of i that equals 1 is i^4 . 4 is the order but not the period of i . So, a complex number which has a finite order need not have a period. However, if n is the period of z , then by definition and (1), $z^{2^n} = z$ and hence

$$z^{2^n-1} = 1 \quad (2)$$

So the number z has a finite order (which may, however, be less than $2^n - 1$ as the order is defined as the exponent of the least power that equals 1). All we can say in general is that the order divides $2^n - 1$.

Coming to complex numbers having period 2, such a complex number z must satisfy $z^4 = z$ and hence

$$z^3 = 1 \quad (3)$$

as $z \neq 0$. There are three such complex numbers, 1, ω and ω^2 . Out of these, 1 has period 1. But the other two have order 3 and period 2 each because $\omega^4 = \omega^3\omega = \omega$ and $(\omega^2)^4 = \omega^8 = \omega^6 \times \omega^2 = \omega^2$.

For $n = 3$ the relevant equation is $z^7 = 1$. There are 7 such complex numbers. They are the seventh roots of unity. Again, 1 has period 1. But we claim that the remaining 6 have period 3. As 7 is a prime, all six of them are primitive 7-th roots of unity. (See Exercises (7.6) and (7.7) for more.) That means no smaller power of any of them can equal 1. Hence there are 6 complex numbers with period 3.

The case $n = 4$ is more interesting. Here, the equation is $z^{15} = 1$. There are 15 roots, viz. the 15-th roots of unity. But they are not all primitive as 15 is not a prime. They can be expressed as powers of $\alpha = e^{2\pi i/15}$. And α^k is a primitive 15-th root of unity if and only if k is relatively prime to 15. Note that $\alpha^5 = \omega$ which has period 2. Similarly $\omega^2 = \alpha^{10}$ has period 2. Excluding these two, we claim that there are

$15 - 1 - 2 = 12$ complex numbers with period 4. Note that α^3 is not a primitive 15-th root because 3 is not relatively prime to 15 either. But still its period is 4 because 4 is the smallest positive integer m such that $(\alpha^3)^{2^m-1} = 1$. (For a direct verification, $\alpha^3 \neq 1$, $(\alpha^3)^7 = \alpha^{21} = \alpha^6 \neq 1$, but $(\alpha^3)^{15} = \alpha^{45} = (\alpha^{15})^3 = 1^3 = 1$.) Indeed, no power of α , has period 3, because if $(\alpha^k)^7 = 1$, then 15 divides $7k$ which is possible only when 15 divides k . But in that case, $\alpha^k = 1$ which has period 1.

The case $n = 5$ is analogous to $n = 3$ because like $2^3 - 1$, $2^5 - 1 = 31$ is also a prime. Hence there are 30 primitive 31-th roots of unity and all of them have period 5. The only remaining root, 1, has period 1.

To handle the general case of finding the complex numbers which have a given period n , we note that if a complex number z_0 has period n , then first of all, $z_0^r = 1$ where $r = 2^n - 1$. The equation $z^r = 1$ has r roots, viz. all the r -th roots of unity. They can be listed as $\alpha, \alpha^2, \dots, \alpha^r$ where $\alpha = e^{2\pi i/r}$ and z_0 is one of them. But not every solution of this equation has period n . Some of them may have periods less than n . For example, as happened when $n = 4$, out of the 15 solutions of $z^{15} = 1$, 1 had period 1 while $\omega (= \alpha^5)$ and $\omega^2 = \alpha^{10}$ had periods 2 each.

Summing up, the general procedure is to take all roots of $z^r = 1$ as potential candidates that may have n as their period and then to discard those among them which have periods less than n . We now show that if $z^r = 1$, where $r = 2^n - 1$, then the period of z is some divisor of n . We go back to the very definition of the period. We are assuming that

$$f^{(n)}(z) = z \quad (4)$$

On the other hand, if d is the period of z then we also have

$$f^{(d)}(z) = z \quad (5)$$

and further d is the smallest positive integer with this property. We claim that d divides n . If not, then by the division algorithm, we can find integers q (for quotient) and remainder e such that

$$n = e + qd \quad (6)$$

where $0 < e < d$. (Normally, the R.H.S. is written as $qd + e$ so as to bring the remainder at the end. But here there is some reason to change the order as will be clear later.) Now,

$$f^{(n)}(z) = f^{(e)}(f^{(qd)}(z)) \quad (7)$$

(Writing the composite of a function f with itself taken n times in the exponential form $f^{(n)}$ is compact and convenient because the associativity of the composite makes the elementary laws of indices hold true. Thus, $f^{(5)} = f \circ f \circ f \circ f \circ f$ and for any z , $f^{(5)}(z) = f(f(f(f(f(z))))))$. This equals $f^{(3)}(f(f(z)))$ and hence $f(f(f(f(f(z))))))$. Therefore $f^{(5)} = f^{(3)} \circ f^{(2)}$. Similarly, $(f^{(2)})^{(3)}$ equals $f^{(6)}$ because for every z , both of them equal $(f \circ f \circ f \circ f \circ f \circ f)(z) = f(f(f(f(f(f(z))))))$. Note that powers of z (except z itself) are nowhere involved!)

Since d is the period of f for z , z is a fixed point of $f^{(d)}$, i.e. $f^{(d)}(z) = z$. But then $f^{(2d)}(z) = f^{(d+d)}(z) = f^{(d)}(f^{(d)}(z)) = f^{(d)}(z) = z$. In other words, z is also a fixed point of $f^{(2d)}$. But then $f^{(3d)}(z) = f^{(d)}(f^{(2d)}(z)) = f^{(d)}(z) = z$. That means z is also a fixed point of $f^{(3d)}$. Continuing, z is a fixed point of $f^{(4d)}, f^{(5d)}, \dots$ and, finally, of $f^{(qd)}$. If we now apply $f^{(e)}$, the R.H.S. of (7) is simply $f^{(e)}(z)$. (This is the reason we wrote the R.H.S. of (6) as $e+qd$ rather than $qd+e$. After all even though the addition of integers is commutative, the composition of functions is not. When we take the composite of two functions, say $g \circ h$, the first function g is to be applied after, and not before h .)

But the L.H.S. of (7) also equals z because even though n may not satisfy the minimality condition (which d does), it still satisfies $f^{(n)}(z) = z$. All put together, we get $f^{(e)}(z) = z$. Hence z is a fixed point of $f^{(e)}$. But, by assumption, $e < d$ contradicting the minimality property of d , viz. that it is the least positive integer d for which z is a fixed point of $f^{(d)}$. This contradiction proves that d divides n .

We record the result just obtained as a theorem.

Theorem 1: Suppose n is a positive integer and z is a complex number in S^1 for which (2) holds. Then z has a period and it is a divisor of n .

Proof: (2) implies that $z^{2^n} = z$. Hence n satisfies the major condition to be the period of z . But it need not satisfy the minimality condition. In that case, let d be the smallest positive integer for which $z^{2^d} = z$ holds. Then by definition, z has period d and by what we proved above d divides n . ■

We shall also need another result which is fairly well known from number theory.

Theorem 2: If d and n are positive integers, then $2^d - 1$ divides $2^n - 1$ if and only if d divides n .

Proof: For the direct implication, assume $n = dk$. Put $u = 2^d$. Then $2^{dk} = (2^d)^k = u^k$. Hence $2^n - 1 = u^k - 1$. As k is a positive integer, we have

$$2^n - 1 = u^k - 1 = (u - 1)(u^{k-1} + u^{k-2} + \dots + u + 1) \quad (8)$$

As u is an integer so is the second factor. Hence $u - 1$ i.e. $2^d - 1$ divides $2^n - 1$.

The proof of the converse is similar to the proof of Theorem 1. We are given that $2^d - 1$ divides $2^n - 1$ and we have to prove that $d|n$. If it does not, write n as

$$n = qd + e \quad (9)$$

where q and e are integers with $1 \leq e < d$. Then

$$2^n - 1 = 2^{qd+e} - 1 = (2^{qd} - 1)2^e + 2^e - 1 \quad (10)$$

Since $d|qd$, $2^d - 1$ divides $(2^{qd} - 1)$ by the direct implication and hence divides the first term on the R.H.S. By assumption, it divides the L.H.S. too. Hence it must divide the second term on the R.H.S. But that is a contradiction because since $0 < e < d$, $1 < 2^e < 2^d$. So we must have $d|n$. ■

It should not be supposed that every divisor of $2^n - 1$ is of the form $2^d - 1$ for some divisor d of n . For example, take $n = 4$. Then $2^4 - 1 = 15$ which has 1, 3, 5 and 15 as divisors. Out of these 1, 3 and 15 are of the form $2^d - 1$ for $d = 1, 2, 4$ respectively. But 5 is not. The theorem above is not applicable for divisors of this form.

For every positive integer n , let $P(n)$ be the number of complex numbers in S^1 which have period n . By direct calculations we proved that

$$P(1) = 1, \quad P(2) = 2, \quad P(3) = 6 \quad \text{and} \quad P(4) = 12 \quad (11)$$

For every positive integer n , there are $2^n - 1$ complex numbers which are potential candidates to have period n , viz. all n -th roots of unity where

$r = 2^n - 1$. They can be explicitly listed as $\alpha, \alpha^2, \alpha^3, \dots, \alpha^r$ where $\alpha = e^{2\pi i/r}$. (11) was obtained by analysing each of these r potential candidates. But the two theorems above give a more efficient method. It is based on complementary counting. Let S_n be the set of all $2^n - 1$ potential candidates to have period n . These are the roots of $z^{2^n-1} = 1$. The only reason some $z_0 \in S_n$ will fail to have n as its period is that it has a smaller period, say d . By Theorem 1, d divides n . Also now z_0 is one of the $2^d - 1$ potential candidates to have period d , viz. all s -th roots of unity where $s = 2^d - 1$. (Actually, z_0 has period d . So it is not just a potential candidate, but a successful candidate. But right now we are counting only potential candidates.) Each of these s elements has a period d or less. But none of them can have period n . So, we can safely discard these $2^d - 1$ elements from S_n , the set of all potential candidates.

Put differently, every rejected candidate, say z_0 , belongs to S_d for some proper divisor d of n . d may vary as z_0 varies. But if we take the union of all S_d 's as d varies over all divisors of n other than n , then we get precisely the set of rejected candidates. Therefore we have

$$\begin{aligned}
P(n) &= |S_n - \bigcup_{d|n, d < n} S_d| \\
&= |S_n| - \left| \bigcup_{d|n, d < n} S_d \right| \\
&= 2^n - 1 - \left| \bigcup_{d|n, d < n} S_d \right| \tag{12}
\end{aligned}$$

where the union is taken over all divisors of n other than n .

When n is a prime, 1 is the only divisor of n and (12) reduces to $P(n) = 2^n - 1 - |S_1| = 2^n - 2$ because $S_1 = \{1\}$. We already verified this for $n = 2, 3, 5$.

When n has a single proper factor, n is of the form p^2 where p is a prime. Now the union is $S_1 \cup S_p$ which equals S_p since $S_1 \subset S_p$. So $P(p^2) = 2^{p^2} - 1 - (2^p - 1) = 2^{p^2} - 2^p$. This tallies with $P(4) = 12$, taking $p = 2$. As a new example, we get $P(9) = 2^9 - 2^3 = 512 - 8 = 504$. A direct calculation would be cumbersome. (More generally, if $n = p^k$, then the proper divisors of n are $1, p, p^2, \dots, p^{k-1}$. The subsets $S_1, S_p, S_{p^2}, \dots, S_{p^{k-1}}$ form an ascending chain and so their union

is simply $S_{p^{k-1}}$. Therefore $P(p^k) = 2^{p^k} - 2^{p^{k-1}}$. In particular, $P(8) = 2^8 - 2^4 = 512 - 16 = 496$.)

The applicability of the formula (12) depends upon the ability to identify all factors of n . There is no easy way to do this. If we know the prime power factorisation of n , then this is easy. But that only passes the buck, because getting the prime power factorisation of an integer is, itself, an extremely difficult task. (If somebody can do that quickly for every integer, most passwords would be vulnerable since they are encoded in terms of factorisations of numbers.)

But if we are given the prime power factorisation of n , then it is not difficult to identify all divisors d of n . The present question asks to find how many complex numbers have period 2025. The fact that we are given $2025 = 3^4 \times 5^2$ as a hint indicates that this is a right approach. Further, the prime power factorisation of 2025 has only two primes, 3 and 5. The following simple results then reduce the computation considerably.

Theorem 3: If $n = p^a q^b$ where p, q are distinct primes and a, b are positive integers, then there exist proper divisors d_1, d_2 of n such that every proper divisor of n divides either d_1 or d_2 (or both).

Proof: We let $d_1 = n/p = p^{a-1} q^b$ and $d_2 = n/q = p^a q^{b-1}$. If $d = p^i q^j$ divides n , but does not equal n then either $i < a$ or $j < b$. In the first case, $d|d_1$ while in the second, $d|d_2$. ■

The two divisors of n are called its **maximal divisors**. Informally, a maximal element of a set is an element which is not the maximum, but next to it, that is, there is nothing between that element and the maximum. (Here ‘between’ is in the sense of divisibility. So, to say that b is between a and c means that a divides b and b divides c .) Thus, the maximum divisor of 36 is 36 itself. It is unique. But 12 is a maximal divisor as there is no divisor of 36 between 12 and 36. 18 is also a maximal divisor. So, in general, maximal divisors are not unique. (More generally, if the prime power factorisation of an integer n involves only the primes $p_1, p_2, \dots, p_r$, then the maximal divisors of n are n/p_i , $1 \leq i \leq r$.) Their relevance in our problem comes from the following result. We confine to the case where $r = 2$.

Theorem 4: Assume n is a positive integer whose prime factorisation consists of powers of only two prime factors. Let d_1, d_2 be the maximal divisors of n . Then

$$P(n) = 2^n + 1 - 2^{d_1} - 2^{d_2} + |S_{d_1} \cap S_{d_2}| \quad (13)$$

Proof: We already know that all the $2^n - 1$ elements of S_n are potential candidates to have period n . If some $z_0 \in S_n$ has period, say d , which is less than n , then d is a proper divisor of n . By the theorem above, either $d|d_1$ or $d|d_2$. If it divides d_1 , then by Theorem 2, $2^d - 1$ divides $2^{d_1} - 1$. Therefore all the roots of $z^{2^d - 1} = 1$ also satisfy $z^{2^{d_1} - 1} = 1$. In other words, $S_d \subset S_{d_1}$. Similarly, if $d|d_2$, then $S_d \subset S_{d_2}$. Therefore $\bigcup_{1 < d < n} S_d \subset S_{d_1} \cup S_{d_2}$. But actually equality holds because d_1 and d_2 are also among the divisors of n that contribute to the union. Thus we have reduced the union appearing in (12) to the union of just two sets.

$$\bigcup_{d|n} S_d = S_{d_1} \cup S_{d_2} \quad (14)$$

(13) follows by applying the principle of inclusion and exclusion to (14) and putting the values $|S_{d_1}| = 2^{d_1} - 1$ and $|S_{d_2}| = 2^{d_2} - 1$. ■

The stage is almost set to calculate $P(2025)$ which is our final goal. All that remains is to calculate the last term in (13). Normally, there is little that can be said, in general, about the cardinality of the intersection of two sets in terms of the cardinalities of those two sets, except perhaps to give trivial upper and lower bounds. To calculate it exactly, we must use the genesis of those two sets, i.e. how they are formed. (Contrast this with the cardinality of the cartesian product of two sets which always equals the product of their cardinalities, regardless of how those two sets were obtained.) In the present case, S_{d_1} is the set of all r_1 -th roots of unity and S_{d_2} is the set of set of all r_2 -th roots of unity where $r_1 = 2^{d_1} - 1$ and $r_2 = 2^{d_2} - 1$. The following result (another well-known theorem from number theory) uses these to express their intersection.

Theorem 5: Suppose r_1 and r_2 are positive integers and their g.c.d. is r . Then a complex number z is both an r_1 -th root of unity and an r_2 -th root of unity if and only if it is an r -th root of unity.

Proof: Suppose a complex number z is an r -th root of unity, i.e. $z^r = 1$. Then $z^{r_1} = 1$ because r divides r_1 . (For details, assume $r_1 = rp$ where p is an integer. Then $z^{r_1} = z^{rp} = (z^r)^p = 1^p = 1$.) Similarly, $z^{r_2} = 1$.

The proof of the converse crucially uses a property of the g.c.d. of two positive integers, viz. that it can be expressed as a linear combination of them (with integer coefficients). So suppose $r = r_1u + r_2v$ where u, v are two (not necessarily positive) integers. For the converse, suppose $z^{r_1} = z^{r_2} = 1$. Then $z^{r_1u} = (z^{r_1})^u = 1^u = 1$. (This holds even if u is negative because powers with negative exponents are powers of the reciprocal and the reciprocal of 1 is 1.) Similarly $z^{r_2v} = 1$. Hence $z^r = z^{r_1u+r_2v} = z^{r_1u}z^{r_2v} = 1 \times 1 = 1$. ■

So we can now say that $|S_{d_1} \cap S_{d_2}| = |S_r|$ where r is the g.c.d. of $2^{d_1} - 1$ and $2^{d_2} - 1$. Although this is a reasonable task when d_1, d_2 are very small, in general $2^{d_1} - 1$ and $2^{d_2} - 1$ are huge numbers even when d_1, d_2 are relatively small (say two digits) numbers. Moreover to find their g.c.d., we need to factorise them and the factors are very unpredictable. For example, the prime power factorisation of $2^{36} - 1$ is

$$2^{36} - 1 = 3^3 \times 5 \times 7 \times 13 \times 19 \times 73 \times 4033 \quad (15)$$

So, in all $2^{36} - 1$ has $4 \times 2^6 = 256$ factors (including 1 and 36). Out of these only 9 can be identified by Theorem 2. They are of the form $2^d - 1$ for $d = 1, 2, 3, 4, 6, 9, 12, 18$ and 36. So, if $d_1 = 36$ and d_2 is some comparable number, then finding the g.c.d. of $2^{d_1} - 1$ and $2^{d_2} - 1$ is a horrendous task. It would be so much nicer if the g.c.d. could be expressed succinctly and preferably directly in terms of the numbers d_1 and d_2 rather than the far bigger numbers $2^{d_1} - 1$ and $2^{d_2} - 1$.

The following theorem, also based on Theorem 2 above, is tailor made for this purpose.

Theorem 6: For any two positive integers d_1 and d_2 , the g.c.d. of $2^{d_1} - 1$ and $2^{d_2} - 1$ equals $2^d - 1$ where d is the g.c.d. of d_1 and d_2 .

Proof: Denote the g.c.d. of $2^{d_1} - 1$ and $2^{d_2} - 1$ by e . We have to prove that $e = 2^d - 1$. We do so by showing that both e and $2^d - 1$ divide

each other. Certainly, since $d|d_1$, Theorem 2 implies that $2^d - 1$ divides $2^{d_1} - 1$. Similarly, $2^d - 1$ divides $2^{d_2} - 1$. As $2^d - 1$ divides both $2^{d_1} - 1$ and $2^{d_2} - 1$, it divides their g.c.d., which is e . So, $2^d - 1$ divides e .

The proof of the converse is structurally similar to but conceptually more complicated than the proof of the converse implication in the last theorem. We first use the same property of the g.c.d. that was used in the proof of Theorem 5 and write d as some linear combination of d_1 and d_2 , say $d = d_1u + d_2v$ where u, v are two (not necessarily positive) integers. Since e divides $2^{d_1} - 1$, we have $2^{d_1} = ke + 1$ for some integer k . Similarly, $2^{d_2} = me + 1$ for some integer m . Therefore,

$$2^d = 2^{d_1u + d_2v} = (2^{d_1})^u (2^{d_2})^v = (ke + 1)^u (me + 1)^v \quad (16)$$

If the integers u, v are positive then we can expand each factor of the R.H.S. by the binomial theorem. Then their product will be some multiple of e plus 1. That is, $2^d - 1$ will be a multiple of e . So $e|(2^d - 1)$ and further $e = 2^d - 1$ as claimed.

But the trouble is that u, v cannot both be positive. If they were, then d would be much less than $d_1u + d_2v$. So, we have to tackle the case where one of them, say u , is a negative integer. (Both u and v cannot be negative either because in that case $d_1u + d_2v$ will be negative too and cannot equal d .)

This difficulty did not arise in the proof of Theorem 5. There z and z^{r_1} were non-zero complex numbers and there is no difficulty in taking the power of a non-zero complex number w with negative integral exponent. If u is a negative integer, then w^u is simply $(1/w)^{-u}$ where $1/w$ is the reciprocal of w . What makes this work is that every non-zero complex number has a reciprocal (a multiplicative inverse, to use a fancy language).

This is where the proof gets conceptually more difficult than that of the converse implication in Theorem 5. We need something that is the equivalent of a multiplicative inverse. But no integers (other than ± 1) have multiplicative inverses w.r.t. the usual multiplication. But the job can be done by taking what is called modular arithmetic. (See Comment No. 15 of Chapter 4 for definitions and some elementary properties.) In the present proof, we work with modulus e , which is the g.c.d. of $2^{d_1} - 1$ and $2^{d_2} - 1$. Then e is, first of all, a common divisor

of both. To say that e divides $2^{d_1} - 1$ is equivalent to saying that $2^{d_1} \equiv 1$ (modulo e), or, in other words, the residue class of 2^{d_1} is the same as that of 1. Compactly, this is expressed by writing $[2^{d_1}]_e = [1]_e$, and still more compactly as $[2^{d_1}] = [1]$. (e is called the **modulus** and is suppressed from the notation when understood.) But $[2^{d_1}]$ is the same as $[2]^{d_1}$, which, in turn equals $[2][2^{d_1-1}]$.

The point to note is that even though the integer 2 has no multiplicative inverse (w.r.t. the usual multiplication), under modulo e multiplication, the residue class $[2]$ has an inverse, viz. $[2^{d_1-1}]$. Similarly, $[2^{d_2-1}]$ is a multiplicative inverse of $[2]$. (It might appear strange that the same element has two inverses. But note that their residue classes modulo e are the same. To see this, suppose $2^{d_1} = 1 + ep$ for some integer p . Then since 2^{d_1} is even, ep and therefore p is odd. Similarly, $2^{d_2} = 1 + eq$ for some odd q . Now, $2^{d_1-1} - 2^{d_2-1} = \frac{2^{d_1} - 2^{d_2}}{2} = \frac{1+ep - (1+eq)}{2} = e \frac{p-q}{2}$. The second factor is an integer since p, q are both odd. Therefore the residue classes modulo e of 2^{d_1-1} and 2^{d_2-1} are equal.)

The proof that e divides $2^d - 1$ can now be completed much the same way as the proof of Theorem 5. The assertion is equivalent to saying that $[2^d]$ equals $[1]$. And this can be done using $d = d_1u + d_2v$ as follows.

$$\begin{aligned} [2^d] &= [2^{d_1u + d_2v}] \\ &= [2^{d_1u}][2^{d_2v}] \end{aligned} \tag{17}$$

$$= ([2^{d_1}])^u ([2^{d_2}])^v \tag{18}$$

$$= ([1])^u ([1])^v \tag{19}$$

$$= [1][1] = [1] \tag{20}$$

As we have proved both $(2^d - 1) | e$ and $e | (2^d - 1)$, the proof is complete.
■

Putting together all the pieces, we get

Theorem 7: For every positive integer n whose prime power factorisation has only two primes,

$$P(n) = 2^n - 2^{d_1} - 2^{d_2} + 2^d \tag{21}$$

where d_1, d_2 are the maximal divisors of n and d is their g.c.d.

Proof: By the last theorem, $S_{d_1} \cap S_{d_2} = S_d$. So $|S_{d_1} \cap S_{d_2}| = |S_d| = 2^d - 1$. The result follows by substituting this into (13). ■

As the simplest illustration, suppose $n = 36$. Then $d_1 = 12, d_2 = 18$ (or vice versa). In either case, $d = 6$. So $P(36) = 2^{36} - 2^{12} - 2^{18} + 2^6$. Just imagine how long it would take to obtain this by using (15) and discarding those potential candidates which have periods 1, 2 or 3.

Finally, in our problem, $n = 2025 = 3^4 \times 5^2$. So $d_1 = 2025/5 = 81 \times 5 = 3^4 \times 5 = 405$ and $d_2 = 2025/3 = 3^3 \times 5^2 = 27 \times 25 = 675$. Also $d = 3^3 \times 5 = 135$. Therefore, $P(2025) = 2^{2025} - 2^{405} - 2^{675} + 2^{135}$. ■

Theorem 7 can be obviously generalised if the prime power factorisation of n has powers of more than two primes. As the simplest such case, assume that n is of the form $p_1^{k_1} p_2^{k_2} p_3^{k_3}$ where p_1, p_2, p_3 are distinct primes and k_1, k_2, k_3 are some positive integers. Then the maximal divisors of n are $d_1 = n/p_1, d_2 = n/p_2$ and $d_3 = n/p_3$. And we have

$$P(n) = |S_n| - (|S_{d_1}| + |S_{d_2}| + |S_{d_3}|) + (|S_e| + |S_f| + |S_g|) - |S_h| \quad (22)$$

where e, f, g are the g.c.d.'s of d_1, d_2, d_3 taken two at a time and h is the g.c.d. of all three. This follows by applying the principle of inclusion and exclusion to the complement of the union of three subsets, S_{d_1}, S_{d_2} and S_{d_3} and using Theorem 6 thrice and also its extension for the case of three divisors.

This is a very challenging problem whose solution requires certain theorems from number theory. Considering the level of the candidates appearing for this examination, we have given not only the proofs of these theorems, but also many explanations and illustrations. But even without them, the solution would be extremely lengthy. Such problems are more suitable as take home assignments or term projects.

- Q. 5 Let a, b, c be non-zero real numbers such that $a + b + c \neq 0$. Assume that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a + b + c}.$$

Show that for any odd integer k ,

$$\frac{1}{a^k} + \frac{1}{b^k} + \frac{1}{c^k} = \frac{1}{a^k + b^k + c^k}.$$

Solution and Comments: The given equation simplifies to

$$(a + b + c)(bc + ca + ab) = abc \quad (1)$$

This equation by itself cannot determine a, b, c . But we note that the terms in it are precisely the elementary symmetric functions in a, b, c . Hence a, b, c are the roots of the cubic equation

$$x^3 - \alpha x^2 + \beta x - \gamma = 0 \quad (2)$$

where

$$\alpha = a + b + c, \beta = bc + ca + ab \text{ and } \gamma = abc \quad (3)$$

Because of (1), we have $\gamma = \alpha\beta$. Hence (2) can be written as

$$x^3 - \alpha x^2 + \beta x - \alpha\beta = 0 \quad (4)$$

Factoring the L.H.S.,

$$(x - \alpha)(x^2 + \beta) = 0 \quad (5)$$

Hence α is a root of the cubic. But already a, b, c are the roots. Hence α must equal one of them. Because of symmetry, we may assume that $\alpha = a$. But that means that $a + b + c = a$, whence $b + c = 0$. Thus the given relation in a, b, c implies that $c = -b$. For an odd integer k , $c^k = -b^k$. Therefore,

$$\frac{1}{a^k} + \frac{1}{b^k} + \frac{1}{c^k} = \frac{1}{a^k} + 0 = \frac{1}{a^k} \quad (6)$$

while

$$\frac{1}{a^k + b^k + c^k} = \frac{1}{a^k + 0} = \frac{1}{a^k} \quad (7)$$

This proves the assertion. ■

A simple, but very tricky problem. The trick is to form the cubic with roots a, b, c and recognise that because of the given relationship between them, $a + b + c$ is also a root.

- Q. 6 Let $\mathbb{N}$ denote the set of natural numbers, and let $(a_i, b_i), 1 \leq i \leq 9$, be nine distinct tuples in $\mathbb{N} \times \mathbb{N}$. Show that there are three distinct elements in the set $\{2^{a_i}3^{b_i} : 1 \leq i \leq 9\}$ whose product is a perfect cube.

Solution and Comments: Since the integers 2 and 3 are relatively prime, the only way a number of the form 2^a3^b where a, b are integers can be a perfect cube is that 2^a and 3^b are perfect cubes. And this happens if and only if a, b are both divisible by 3.

For any three distinct indices i, j, k , the product $2^{a_1}3^{b_1}2^{a_2}3^{b_2}2^{a_3}3^{b_3}$ equals $2^{a_1+a_2+a_3}3^{b_1+b_2+b_3}$. From the observation above, this will be a perfect cube if and only if $a_1 + a_2 + a_3$ and $b_1 + b_2 + b_3$ are both divisible by 3.

So, given 9 such pairs (a_i, b_i) , our job is to show that there exist some distinct indices i, j, k such that $a_1 + a_2 + a_3$ and $b_1 + b_2 + b_3$ are both divisible by 3. Note that the actual numbers do not matter so much as their residue classes modulo 3. There are 3 such residue classes, 0, 1 and 2. Let us denote the set of the a_i 's by A and the set of b_j 's by B . The statement of the problem does not make it clear in what sense the word 'tuple' is used. We take it to mean an ordered pair. Two ordered pairs (a_i, b_i) and (a_j, b_j) are considered equal if both $a_i = a_j$ and $b_i = b_j$. So, it may happen that $a_i = a_j$ but $b_i \neq b_j$. Even then the pairs (a_i, b_i) and (a_j, b_j) would be considered distinct. If this happens, we replace a_j by another integer which is different from a_i but is in the same residue class as a_j . This does not change the problem because as observed earlier, the conclusion depends only on the residue classes of the numbers and not on the numbers *per se*. We similarly change one of the two b 's in case they happen to be equal.

We can now suppose that the sets $A = \{a_1, a_2, \dots, a_9\}$ and $B = \{b_1, b_2, \dots, b_9\}$ have 9 elements each. We define a bijection $f : A \rightarrow B$ by $f(a_i) = b_i$ for $1 \leq i \leq 9$. We may call a_i and b_i as mates of each other. (Note that f is not necessarily residue class preserving. That is, an element and its mate need not always have the same class, they may belong to different residue classes. In fact, if f preserved residue classes, the problem would be trivial. All we need to do is to choose any three a 's belonging to the same residue class (whose existence is guaranteed if we have at least seven a 's as there are only three residue classes) and take their mates.)

The crucial observation on which the solution is based is that for any three (distinct) integers, their sum is divisible by 3 (i.e. its residue class modulo 3 is 0) if and only if either all three are in the same residue class or they are all in different residue classes. The proof is by trying various combinations of residue classes.

There is a colourful way to state this observation which makes it more appealing, if only verbally. Assign arbitrary (but different) colours to the residue classes. Then the sum of the elements of a triple of three a 's (or of three b 's) is divisible by 3 if and only if the triple is either monochromatic or trichromatic. (The one type left is dichromatic, in which two members of the triple occupy one residue class and the remaining one some other.) Although residue classes and colouring classes mean the same, generally we shall use the former term for A and the latter for B .

Before proving any results, we remark that the data as well as the conclusion are perfectly symmetric in A and B . So, if we prove the existence of a desired triple under some assumptions on A (i.e. on the a 's), then a similar result will hold if we make the same assumptions on B (and its elements). This is called the principle of **duality**. Its analogues figure in various branches of mathematics. For example, if some result is proved about increasing functions then there is the dual result about decreasing functions. (As a concrete example, if every monotonically increasing sequence which is bounded above is convergent, then every monotonically decreasing sequence which is bounded below is also convergent.)

As immediate consequences of the simple observations we have made so far, we have the following results.

Theorem 1: If any five (or more) a_i 's are in the same residue class, then no matter how the b 's are distributed, there is a triple of the desired type. By duality, the same holds if 5 or more b 's are in the same residue class.

Proof: Without loss of generality, suppose a_1, a_2, a_3, a_4, a_5 all belong to the same residue class. Then the sum of any three of them is divisible by 3. Look at the corresponding b 's, b_1, b_2, b_3, b_4, b_5 . If any three of them, say b_1, b_2, b_3 , are in the same residue class (not necessarily the

same residue class to which the five a 's belong), then these three, along with their mates will form a desired triple. So, suppose no three of b_1, b_2, b_3, b_4, b_5 are in the same residue class. The only way this can happen is if two of them are in one residue class, two in some other and the remaining one in the third. But then picking one from each class, they and their mates in A form a desired triple. ■

We can paraphrase this result by saying that if A_1 is a monochromatic subset of A and $|A_1| \geq 5$, then there exists a desired triple in which all a 's are in A_1 .

The next two theorems have weaker hypotheses and hence less assertive. Instead of guaranteeing a desired triple, they tell what happens if such a triple does not exist. We prefer to state them in the language of colours rather than residue classes.

Theorem 2: Suppose A_1 is a monochromatic subset of A with $|A_1| = 4$. Then either there exists a desired triple (with its a 's in A_1) or $f(A_1)$ is dichromatic with two elements of each colour.

Proof: Again, suppose without loss of generality that $A_1 = \{a_1, a_2, a_3, a_4\}$. If $\{b_1, b_2, b_3, b_4\}$ is also monochromatic, we are done because we take any three a 's in A_1 and their mates. If $f(A_1)$ is trichromatic then also we are done. We choose representatives from each of these colours. They and their mates in A_1 form a desired triple. The remaining possibility is that $f(A_1)$ is dichromatic with two colours, say red and blue. Here, too, if three b 's are of the same colour, then they along with their mates form a desired triple. The only remaining case is when $f(A_1)$ is dichromatic with two elements of each colour. ■

Theorem 3: Suppose A_1 is a monochromatic subset of A with $|A_1| = 3$. Then either there exists a desired triple (with its a 's in A_1) or $f(A_1)$ is dichromatic with two elements of one colour and one of the other.

Proof: This is very similar to the last theorem. If $f(A_1)$ is monochromatic or trichromatic, we are done. Otherwise $f(A_1)$ is dichromatic but all three of its elements cannot be of the same colour (as otherwise it would be monochromatic). ■

As with Theorem 1, Theorems 2 and 3 have dual versions with b 's and a 's interchanged.

With these basic preliminaries, we now turn to the problem. There are infinitely many positive integers and hence infinitely many ways to form the set A . Similarly for B . But as said before, it is not so much the numbers but their residue classes modulo 3 that are relevant to us. As there are three possible residue classes for every positive integer, in all the set A can be formed in 3^9 ways. Same holds for B . So, in theory the problem is finitistic and can be solved by exhibiting a triple of the desired type for every ordered pair of two sequences of length 9 whose entries are modulo 3 residue classes of the elements. But there are $3^9 \times 3^9 = 3^{18}$ such ordered pairs. This is a very huge number and so this method is hardly practicable.

Instead, we classify the possibilities according to how many members of A belong to each residue class. Number the residue classes in an arbitrary but fixed manner. Let A_1, A_2, A_3 be the subsets of A consisting of elements belonging to the first, the second and the third residue classes respectively. Changing the numbering of the residue classes if necessary, we may suppose that $|A_1| \geq |A_2| \geq |A_3|$. We call $(|A_1|, |A_2|, |A_3|)$ a residue partition (or a chromatic partition) of A . Clearly, its entries always add to 9. Because of Theorem 1, if $|A_1| \geq 5$, then already we have a triple of the desired type. So we consider only those partitions in which no entry exceeds 4. There are only three such partitions: (i) (4,4,1), (ii) (4,3,2) and (iii) (3,3,3). We prove the assertion in each case separately. Before doing so, we remark that we can always permute the a 's anyway we like. If we also permute their mates correspondingly, then the conclusion remains unaffected. (To elaborate, suppose (a_1, a_2, a_3) is a desired triple (i.e. $a_1 + a_2 + a_3$ and $b_1 + b_2 + b_3$ are both multiples of 3), and after a permutation of the subscripts, a_1, a_2, a_3 become some a_i, a_j, a_k respectively, then (a_i, a_j, a_k) is a desired triple, provided b_1, b_2, b_3 become b_i, b_j, b_k respectively.)

We now prove the result for each of the three possible chromatic partitions.

(i) **[(4,4,1) Partition.]** Because of the notational simplification mentioned above, we may suppose $A_1 = \{a_1, a_2, a_3, a_4\}$, $A_2 = \{a_5, a_6, a_7, a_8\}$ and $A_3 = \{a_9\}$. We apply Theorem 2 to both A_1 and A_2

and get that either there is a desired triple or both $f(A_1)$ and $f(A_2)$ are dichromatic with two b 's of each colour. But in all there are only three colours. So, one and possibly both of these colours are common.

If the first possibility holds, suppose without loss of generality, that b_1, b_2, b_5, b_6 are red (the common colour), b_3, b_4 are blue and b_7, b_8 are green. The residue class of a_9 is different from all other a 's by assumption. But the colour of b_9 has to be either red, blue or green as these are the only colours. If b_9 is red, then $\{b_1, b_2, b_5, b_6, b_9\}$ is a red quintuplet and by the dual version of Theorem 1, there is a triple of the desired type.

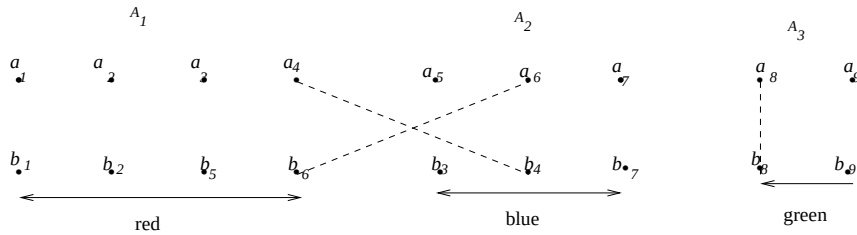
The other two possibilities, viz. b_9 blue and b_9 green are symmetric w.r.t. each other. So, suppose b_9 is blue. Now the situation is that b_1, b_2, b_5, b_6 are red, b_3, b_4, b_9 are blue and b_7, b_8 green. Then both the sets $\{a_1, a_7, a_9\}$ and $\{b_1, b_7, b_9\}$ are trichromatic. Hence $(a_1, b_1), (a_7, b_7), (a_9, b_9)$ is a triple of the desired type.

We still have to consider the case where $f(A_1)$ and $f(A_2)$ are dichromatic with the same two colours, say red and blue appearing in both. In that case, we may assume that b_1, b_2, b_5, b_6 are red and b_3, b_4, b_7, b_8 are blue. If the colour of b_9 is either red or blue, then again we have five b 's of the same colour and by the dual part of Theorem 1, we get a triple of the desired type. Suppose, however, that b_9 is of some third colour, say green. Then a_1, a_7, a_9 and their mates form a desired triple because both are trichromatic triples.

We have thus shown that a triple of the desired type exists in every case where the set of the 9 a 's has a residue partition of the type $(4, 4, 1)$. By duality, the same holds if B has a chromatic partition of the $(4, 4, 1)$ type. Before passing to the next partition, we remark that in the last part of the discussion above, the pair (a_9, b_9) was vital to the desired triple. That shows that 9 is the least integer for which the assertion of the problem is correct. If we have only 8 pairs, a desired triple may not necessarily exist.

(ii) [(4,3,2) Partition.] We begin with $A_1 = \{a_1, a_2, a_3, a_4\}$, $A_2 = \{a_5, a_6, a_7\}$ and $A_3 = \{a_8, a_9\}$. The analysis and break-up into various types is similar to the $(4, 4, 1)$ partition of A , except that instead of Theorem 1, we apply Theorems 2 and 3 to A_1 and A_2 respectively. So, if there is no triple of the desired type with a 's in A_1 or in A_2 , then $f(A_1)$

and $f(A_2)$ are both dichromatic with each colour appearing twice for $f(A_1)$ and one colour twice and the other once for $f(A_2)$. As there are only three colours, one or both the colours are common. First assume that both the colours are common. Call them as red and blue. Then in $f(A_1)$ each appears twice. But in $f(A_2)$, either two are red and the remaining one blue or vice versa. By symmetry, we may assume that in $f(A_2)$, two are red and one blue. So, without loss of generality, suppose b_1, b_2, b_5 and b_6 are red and b_3, b_4 and b_7 are blue. Now consider the colours of the remaining two b 's, viz. b_8 and b_9 . If at least one of them is red, we would have five red b 's and so by Theorem 1 (or, rather its dual) there will be a desired triple. If both of them are blue then we would have five blue b 's and hence a desired triple. The only case left is when at least one of them, say b_8 , has a different colour, say green. Now, the only remaining b is b_9 and it can be blue or green. (Red is already ruled out.) If b_9 is blue, then the chromatic partition of B is $(4, 4, 1)$. This is the dual of the $(4, 4, 1)$ partition of A , which we have already covered. The only remaining case is that b_9 is also green (along with b_8). Now the chromatic partition of B is $(4, 3, 2)$ with which we are still working. Nevertheless, we can find a desired triple by inspection. All we have to do is to find some three a 's which represent three different residue classes so that their mates have different colours. This can be done visually by looking at the following diagram in which the upper row lists the residue classes of a 's and the lower one lists their colour groups.

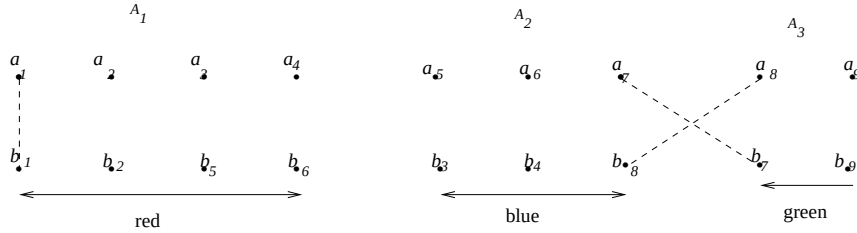


From the diagram we see that a_4, a_6, a_8 and their mates b_4, b_6, b_8 constitute a desired triple. (The selected pairs are shown by joining their members by dotted lines. Of course, many other choices are possible, e.g. a_1, a_7, a_9 and their mates. Note, however, that in the situation pictured in this diagram, exactly one out of a_8 and a_9 has to be chosen.

The reader is encouraged to verify the other examples given so far in the light of such diagrams.)

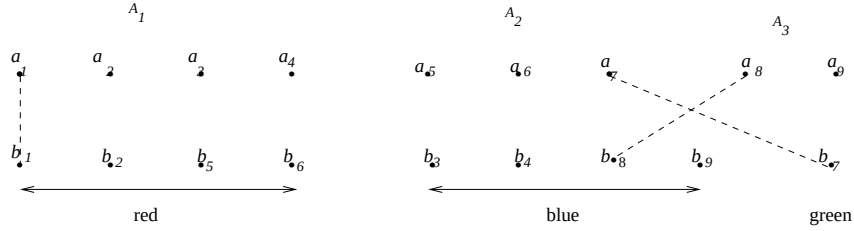
So far, in the discussion of the case where the residue partition of A is of the $(4, 3, 2)$ type, we assumed that both $f(A_1)$ and $f(A_2)$ are dichromatic with the same two colours, viz. red and blue, appearing in both. We now deal with the case where only one of these two, say the red, colour is common. In that case, some third colour, say green, must also appear in $f(A_2)$. Moreover, since $f(A_2)$ has only three elements two cases arise depending upon whether red is the major colour (i.e. appears twice) or a minor colour (i.e. appears only once). We handle them separately. In both of them, we continue with the assumption that $f(A_1)$ is dichromatic with two red and two blue elements. Also, without loss of generality we assume that b_1, b_2 are red and b_3, b_4 are blue as before.

Now, if red is the major colour in $f(A_2)$, then without loss of generality assume that b_5 and b_6 are red. Then b_7 has colour green. In $f(A_3)$, there are two elements, viz. b_8 and b_9 . If at least one of them is red, we are done because now there are 5 red elements in B . If both are blue, then also we are done because now the chromatic partition of B is of the $(4, 4, 1)$ type (with four red, four blue and one green element) whose dual has been handled before. The cases where both b_8 and b_9 are blue or one of them (say b_8) is blue and the other green remain. Unfortunately, in both of them the chromatic partition of B is of the $(4, 3, 2)$ type whose analysis is still going on. It is not time yet to rely on it. Nevertheless, in both cases, we can directly exhibit desired triples by looking at suitable diagrams. When b_8 is blue and b_9 green, the diagram is



from which we see immediately that a_1, a_7 and a_8 along with their mates form a desired triple. In the case where both b_8 and b_9 are blue, we

have 4 red b 's (viz. b_1, b_2, b_5 and b_6), 4 blue b 's (viz. b_3, b_4, b_8, b_9 and 1 green (viz. b_7). So, the chromatic partition of B is of the $(4, 4, 1)$ type. We handled this (or, rather its dual) above. So in this case there is a triple of the desired type. (In a particular case, it is, of course, easier to get a triple directly by looking at the diagram. In the present case, the relevant diagram is



from which we easily see that a_1, a_7, a_8 and their mates form a desired triple. This is the same as the one above. But that is a coincidence.)

Finally, we dispose off the case where $f(A_1)$ is dichromatic with two red and two blue elements and $f(A_2)$ is also dichromatic but with only one red. Without loss of generality suppose b_5 is the only red element in $f(A_5)$. Then the other two elements b_6 and b_7 must be green. So far we have three reds, 2 blues and two greens. The remaining two, b_8 and b_9 , can have any two colours from red, blue and green. As there are three choices for each, in all the two of them can be assigned two colours in 9 different ways. If both are red, then we have a quintuple of reds and we are done. In all other cases, things are far from conclusive. Both blue or both green lead to a chromatic partition of B of the $(4, 3, 2)$ type which we are still working on. Only one red also leads to this type of partition. Finally, one blue and one green give a $(3, 3, 3)$ type chromatic partition of B which we have not yet discussed.

So, no work we have done so far helps us identify a desired triple in any of these eight cases. It has to be found by inspection. Instead of drawing a diagram and showing the desired triple by three dotted lines in each case, we make a table for these eight cases in which in the second column we indicate the assigned colours (abbreviated as r, b or g) of b_8 and b_9 (in that order), the resulting chromatic partition of B in the next three columns (listing elements having red, blue and green colours) and, finally, a desired triple in the last column.

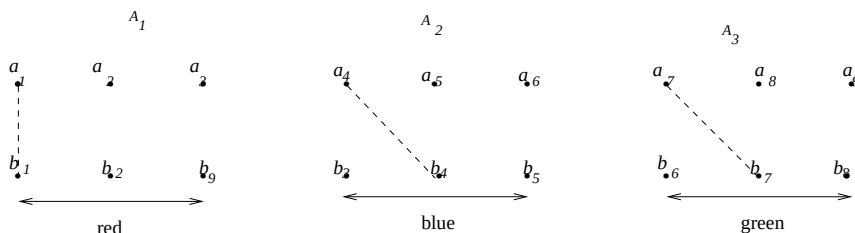
Preparing the table is not all that clerical as it may seem. We already know that b_1, b_2 and b_5 are red, b_3, b_4 are blue and b_6, b_7 are green. It is only the colours of b_8 and b_9 that need to be added. Moreover, a desired triple appearing in the last column can be found even without a diagram, if we keep in mind that $A_1 = \{a_1, a_2, a_3, a_4\}$, $A_2 = \{a_5, a_6, a_7\}$ and $A_3 = \{a_8, a_9\}$. So, to make $\{a_i, a_j, a_k\}$ trichromatic, i should be in the lower range 1 to 4, j in the middle range 5 to 7 and k in the top range 8 and 9. Further b_i, b_j and b_k should have different colours. This requires a little trial. As a general rule, choose in order of increasing freedom. In the second row, for example, we take $k = 9$ and then $j = 7$. Then b_7 and b_9 already have different colours, viz. blue and green. For i there are four choices, 1 to 4. Choose it so that b_i is red. $i = 1$ and $i = 2$ both fit the table.

Ser. No.	(b_8, b_9) colours	Red class	Blue class	Green class	Triple (i, j, k)
1	(r, r)	$\{b_1, b_2, b_5, b_8, b_9\}$	$\{b_3, b_4\}$	$\{b_6, b_7\}$	Thm. 1
2	(r, b)	$\{b_1, b_2, b_5, b_8\}$	$\{b_3, b_4, b_9\}$	$\{b_6, b_7\}$	(1,7,9)
3	(b, r)	$\{b_1, b_2, b_5, b_9\}$	$\{b_3, b_4, b_8\}$	$\{b_6, b_7\}$	(1,7,8)
4	(r, g)	$\{b_1, b_2, b_5, b_8\}$	$\{b_3, b_4\}$	$\{b_6, b_7, b_9\}$	(4,5,9)
5	(g, r)	$\{b_1, b_2, b_5, b_9\}$	$\{b_3, b_4\}$	$\{b_6, b_7, b_8\}$	(4,5,8)
6	(b, b)	$\{b_1, b_2, b_5\}$	$\{b_3, b_4, b_8, b_9\}$	$\{b_6, b_7\}$	(1,7,8)
7	(b, g)	$\{b_1, b_2, b_5\}$	$\{b_3, b_4, b_8\}$	$\{b_6, b_7, b_9\}$	(1,7,8)
8	(g, b)	$\{b_1, b_2, b_5\}$	$\{b_3, b_4, b_9\}$	$\{b_6, b_7, b_8\}$	(1,7,9)
9	(g, g)	$\{b_1, b_2, b_5\}$	$\{b_3, b_4\}$	$\{b_6, b_7, b_8, b_9\}$	(4,5,8)

We have now fully exhausted all possibilities for a $(4, 3, 2)$ residue partition of A . In fact, we have now we have completely settled all cases where at least one residue class in A has 4 or more elements. By duality, the same holds for colour classes of B . This will pay off in settling the only remaining case of possible partitions of A .

(iii) [(**3,3,3**) **Partition.**] As usual, we suppose, without loss of generality that $A_1 = \{a_1, a_2, a_3\}$, $A_2 = \{a_4, a_5, a_6\}$ and $A_3 = \{a_7, a_8, a_9\}$. We apply Theorem 3 to all three. Then in case there is no desired triple coming from any of them, $f(A_1)$, $f(A_2)$ and $f(A_3)$ are dichromatic. In all there are nine elements in B but only three colours. If some colour

appears 4 or more times then we are done as just observed. The only case left is when every colour appears exactly three times. But since $f(A_1), f(A_2)$ and $f(A_3)$ are all dichromatic, the only way each of the three colours, say red, blue and green, appears thrice is if the two colours of the three subsets appear cyclically. Without loss of generality, assume that in $f(A_1)$ red appears twice and blue once, in $f(A_2)$, blue appears twice and green once and in $f(A_3)$ green appears twice and red once. We may also further suppose that b_1, b_2 are red and b_3 blue, in $f(A_2)$ b_4, b_5 are blue and b_6 green and finally, in $f(A_3)$, b_7, b_8 are green and b_9 red. As done several times earlier, we represent this with a diagram and get a desired triple from it.



From the diagram we see at one that $(1, 4, 7)$ is a desired triple.

Thus in every possible case, we have proved the existence of a triple (i, j, k) such that $2^{a_i+a_j+a_k}3^{b_i+b_j+b_k}$ is a perfect cube. ■

The number theory part needed in the solution is minimal. The noteworthy feature pivotal to the entire solution is that the sum of three residue classes modulo 3 equals 0 only in the two extremes, viz. all are equal or no two are equal. This will not be the case if we replace 3 by some other modulus.

The rest of the problem is combinatorial and uses the pigeon hole principle at many places. This principle itself little more than common sense and so is rarely named explicitly when used. But when ingeniously applied it works wonders. A classic result of this type is a theorem of **Ramsey** which says that if there are six persons at a party then either there are three among them, every two of whom know each other or there are three no two of whom know each other. (See the last paragraph of Comment No. 16 of Chapter 6. There is a whole book on Ramsey Theory by Graham, Rothschild and Spencer.)

The present problem resembles the Ramsey problem at least superficially. The solution given above consists of classifying all possible cases of data (whose number is huge, 2^{18} , as remarked earlier) into several classes in such a way that the same reasoning applies to all cases in the same class. But usually, each class runs into several subclasses. That is why the solution is so lengthy. It would be interesting if there is a short, tricky solution similar to the solution of the Ramsey problem.

Fortunately, one such solution is indeed possible and we present it here. It is based on an idea proposed by Agnibesh. As mentioned earlier, in essence the pigeon hole principle merely says that a function whose domain is bigger than its codomain cannot be one-to-one, i.e. takes at least two distinct points to the same point, or equivalently, at least one point in the codomain has at least two preimages. In a slightly more general form (given at the start of Comment No. 16 of Chapter 6), if k is a positive integer such that $|X| > k|R|$ (where X is the domain and R is the range of the function, then at least one element of R has $k + 1$ or more preimages under the function.

In applications, art often lies in choosing this function ingeniously. Recall that we reduced the problem to a setting where $A = \{a_1, a_2, \dots, a_9\}$ and $B = \{b_1, b_2, \dots, b_9\}$ are sets each with 9 distinct elements. Then on each A and B we defined the mod 3 residue class function. That is, for any $a \in A$, $f(a)$ is the mod 3 residue of a and similarly, for elements of B . We proved some simple theorems with hypotheses of the type where some residue class appears 5, 4 or 3 times (i.e. has 5, 4 or 3 preimages). To apply these theorems we had to consider the residue partitions (or chromatic partitions) of A and of B (by duality). Separate arguments were needed to handle the cases of different partitions. As a result, the solution became very lengthy.

In effect, we defined the residue class function on a set whose domain was $A \cup B$. The novel feature of Agnibesh's approach is that it defines a function, say g , on the set $X = \{(a_i, b_i) : 1 \leq i \leq 9\}$ of the given 9 tuples and assigns to a typical element (a_i, b_i) of X the ordered pair $(f(a_i), f(b_i))$ of residue classes modulo 3. That is, $g(a_i, b_i) = (f(a_i), f(b_i))$ for $i = 1, 2, \dots, 9$. We may call g the **joint residue class function**. Here the codomain is the set of 9 possible residue pairs modulo 3, viz. $(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1)$ and $(2, 2)$. For example, if $(37, 15)$ is one of the 9 given tuples, then

$$g(37, 15) = (1, 0).$$

This may sound little more than book keeping. But the advantage is that the codomain set, say $Y = \{0, 1, 2\} \times \{0, 1, 2\}$ can be pictured as a 3×3 grid of squares named A, B, C, D, E, F, G, H and I as shown in the figure below.

(0,0) <i>A</i>	(0,1) <i>B</i>	(0,2) <i>C</i>
(1,0) <i>D</i>	(1,1) <i>E</i>	(1,2) <i>F</i>
(2,0) <i>G</i>	(2,1) <i>H</i>	(2,2) <i>I</i>

This geometric interpretation allows us to talk in terms of rows, columns and diagonals. They are all collinear sets. Collinearity translates into something relevant to our problem. Suppose, for example, that $g(a_1, b_1), g(a_2, b_2)$ and $g(a_3, b_3)$ occupy the squares A, B, C in the first row (not necessarily in that order). This means $f(b_1) = f(b_2) = f(b_3)$, and at the same time $f(a_1), f(a_2)$ and $f(a_3)$ are distinct. As a result, $f(a_1) + f(a_2) + f(a_3)$ and $f(b_1) + f(b_2) + f(b_3)$ both vanish. But that means that the triple $\{(a_1, b_1), (a_2, b_2), (a_3, b_3)\}$ is a desired triple, i.e. where the product

$2^{a_1+a_2+a_3} 3^{b_1+b_2+b_3}$ is a perfect cube. By a very similar argument, if the images (under g) of some three pairs in X occupy any other row, column or diagonal, then we get a desired triple. This can be worded slightly differently. Let R be the range of the function g . Then R consists of those joint residue classes in Y that have at least one preimage (under g) in X . Our argument shows that if some three elements of R are collinear, then by taking one preimage of each one of them, we get a desired triple.

Note, however, that there are non-collinear squares too, which, if present in R will give a desired triple. Consider, for example, the corner square A along with the squares F and H , which are symmetrically located about the diagonal through A but are not adjacent to A . The sum of the

joint residue classes of A, F, H is the joint residue class $(0+1+2, 0+2+1)$ which is simply $(0, 0)$ (modulo 3). Hence by picking one preimage each of A, F and H we get a desired triple. By symmetry of the grid, instead of A we can take any corner and get a triple of non-linear squares whose joint residue classes add to $(0, 0)$. Specifically, each of the non-linear triples $(C, D, H), (G, B, F)$ and (I, D, B) gives rise to a desired triple from the set X as can be checked by a direct computation. Let us call each of these four triples of non-linear squares as corner type triples. All these 12 triples of squares leading to desired triples from the set $X = \{(a_i, b_i) : 1 \leq i \leq 9\}$ are shown in the figure next page.

Triplets that sum up to $(0, 0)$ modulo 3

Collinear Triplets

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

Non-collinear Triplets

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

(0, 0)	(0, 1)	(0, 2)
(1, 0)	(1, 1)	(1, 2)
(2, 0)	(2, 1)	(2, 2)

Thus the fate of the problem depends on the range set R of the joint residue class function $g : X \rightarrow Y$. Nothing is given about the set X . It has 9 elements. So R can have any number of elements between the extremes 1 (when g is a constant function) to 9 (when g is a one-to-one and hence a bijective function (since both X and Y have 9 elements each)).

It might appear at first sight that we need a large range R in order to be able to get at least one collinear or at least one corner type triple contained in R . Curiously, even when $|R|$ is small, we can get a desired triple because of a different reason. Specifically, suppose $|R| \leq 4$. Then since the domain set of g , viz. X has 9 elements, by the pigeon hole principle (or rather, by its minor extension mentioned earlier), at least one element of R has at least 3 preimages (as otherwise, there will be at most $4 \times 2 = 8$ elements in X , a contradiction). Let us suppose that $(a_i, b_i), (a_j, b_j)$ and (a_k, b_k) are three distinct tuples in X which have the same image under the joint residue class function g . This means $f(a_1) = f(a_2) = f(a_3)$ modulo 3 and $f(b_1) = f(b_2) = f(b_3)$ modulo 3. Hence we have $f(a_1) + f(a_2) + f(a_3) = 0$ modulo 3 and $f(b_1) + f(b_2) + f(b_3) = 0$ modulo 3. But then, $(a_i, b_i), (a_j, b_j)$ and (a_k, b_k) together form a desired triple.

The case when $|R| \geq 5$ is more complicated. It amounts to showing that if R is a set consisting of the grid elements, viz., the squares $A, B, C, D, E, F, G, H, I$ then either R contains some triplet of three collinear squares (i.e. rows, columns or diagonals) or some corner type triple consisting of some corner square along with two squares symmetrically located w.r.t. the diagonal passing through the corner but are not adjacent to it (e.g. the triplet (A, F, H)).

So, suppose R contains exactly 5 of the squares A to I , but contains no three collinear squares. We claim that R must contain some corner type triple. We do this by resolving into several cases depending on how many corners of the grid are present in R . This number can vary from 0 to 4 and we dispose these cases separately, not in serial order but rather, in order of ease! We shall frequently refer to the set of the five non-corner elements of the grid. It is the union of the middle row and the middle column. We shall call this set the **cross** because of its familiar shape. It is shown in the diagram with a thickened boundary.

If R contains all four corners, then it has room for only one more square and no matter which one it is, it lies between two corners and thereby gives three collinear squares in R .

At the other extreme, if R has no corner squares in it, then the five elements in R form the 'cross', i.e. the union of the middle row and the middle column, both of which are linear.

Next suppose R contains only one corner square, which we may suppose, because of symmetry, to be the square A . The remaining four squares in R cannot be from the remaining three corners, viz. C, G and I . So the remainder of R consists of the same 'cross', with only one square removed. If the missing square is B, D or E , then A, F, H form a corner type triple contained in R . If F is missing from R , then the middle column is contained in R , while if the missing square is H , then the middle row is contained in R . In both the cases, we get a contradiction.

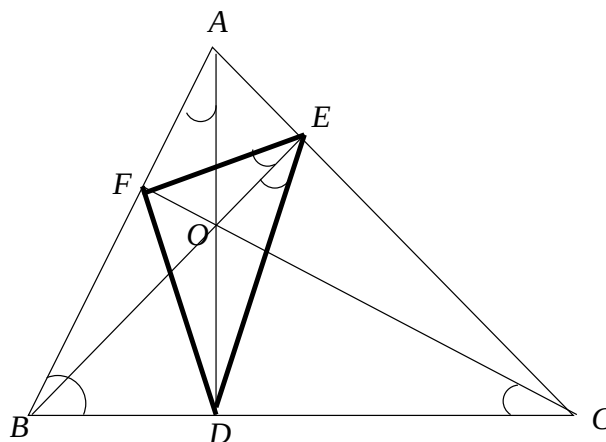
Now suppose that R contains three corners. Again, we may suppose, without loss of generality that they are A, C and I . The remaining corner G is already outside R . As $|R| = 5$, the remaining two elements of R come from the cross. If B, F or E is in R , then R contains a row or a column or a diagonal respectively, a contradiction. The only remaining possibility is that both D and H belong to R . But in that case R contains the corner type triple $\{C, D, H\}$.

Finally, we consider the case where R contains exactly two corner squares. This case is a bit thorny because we have to consider two subcases depending upon whether these two corners are adjacent or diagonal. In the first subcase, assume that the corners A and C are in R , but G and I are not. Then the remaining three squares belonging to R must come from the cross. But B cannot be one of them, as otherwise R contains a row. So, out of the squares D, E, F and H , exactly three are in R . Presence of D, E, F in R creates a contradiction. In all the remaining cases, R contains one or two of the corner type triples (A, F, H) and (C, D, H) . Suppose now that R contains only two diagonally opposite squares, say A and I . This time neither C nor G is in R . The remaining three squares in R must come from the cross. Again, presence of E is ruled out as it would imply that R contains a diagonal. In each of the remaining three cases, R contains one or both

of the corner type triples (A, H, F) and (C, D, H) .

Summing up, we have exhausted all possibilities and shown the presence of a desired triple of the form $\{(a_i, b_i), (a_j, b_j), (a_k, b_k)\}$ in all cases. It may be argued that this method also runs into a large number of cases and subcases. But the advantage over the earlier method is that identifying a desired triple in each case is much simpler and systematic with the help of a single diagram of the grid $\{0, 1, 2\} \times \{0, 1, 2\}$ whose elements are the nine joint residue classes, marked with squares labeled from A to I .

- Q. 7 Consider a ball that moves inside an acute angled triangle along a straight line, until it hits the boundary, which is when it changes direction according to the mirror law, just like a ray of light (angle of incidence = angle of reflection). Prove that there exists a triangular periodic path for the ball, as pictured below.



Solution and Comments: The wording of the problem is confusing. Once the initial position of the ball and its initial direction of motion is fixed, its path is completely determined. The way the problem is worded, it is not clear whether the ball is placed at a given point and we are only given the freedom to choose the direction in which it starts moving or whether we are free to choose the initial location and also the initial direction of its motion. We assume the latter is the case. In that case, the problem is merely to show that there is some inscribed

triangle which has the property that at each of its vertex, the two sides meeting at it are equally inclined to the side of the original triangle on which the vertex lies.

So, let ABC be the original triangle and DEF the desired inscribed triangle, with D, E, F lying on the sides BC, CA, AB respectively. The lines DE and FE are equally inclined to the side CA if and only if they are equally inclined to the normal to CA and hence equally inclined to the direction of the altitude through B . This suggests that if we take D, E, F as the feet of the three altitudes of $\triangle ABC$, then the triangle DEF may have the desired property.

This guess turns out to be correct. Let O be the orthocentre of the triangle ABC , i.e. the point of concurrence of the altitudes AD, BE and CF . Then the quadrilateral $OFAE$ is cyclic as the opposite angles $\angle OFA$ and $\angle OEA$ are both right angles. Hence $\angle OAE = \angle OEF$, being angles subtended by the chord FO at the two points A and E in an arc of the circle passing through O, F, A, E . Hence

$$\angle FEO = \angle FAO \quad (1)$$

Similarly, because of concyclicity of O, E, C, D , we have

$$\angle OED = \angle OCD \quad (2)$$

Since the triangles BDA and BFC are right angled at D and F respectively, we have

$$\angle BAD = 90^\circ - \angle B \quad (3)$$

$$\text{and } \angle FCB = 90^\circ - \angle B \quad (4)$$

which together imply that

$$\angle FAO = \angle FAD = \angle FCD = \angle OCD \quad (5)$$

(1), (2) and (5) together imply that the altitude B bisects the angle $\angle FED$. Hence a ball starting from any point on FE and going along FE will, after reflection in the side AC , move along ED . The same reasoning shows that after being reflected by the side BC it will move along DF and thereafter, when reflected at F will move along FE .

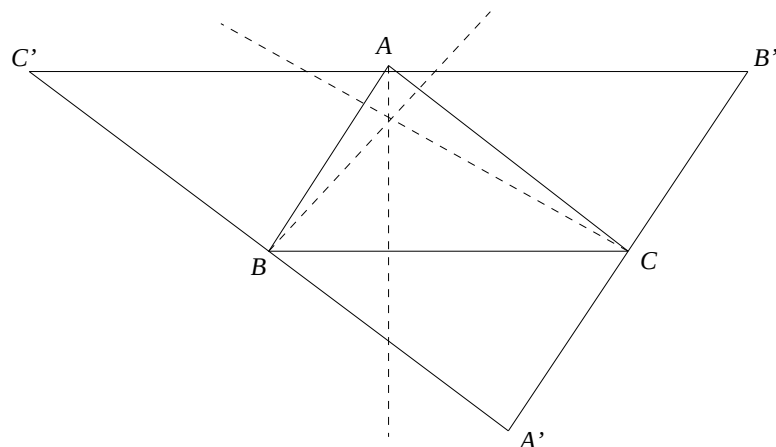
And this cycle will continue. Therefore $F - E - D - F - E - D - F - E - D - F \dots$ is a desired cyclic path for the ball. ■

The triangle DEF is called the **orthic triangle** of the triangle ABC . It is the pedal triangle of the orthocentre of the triangle ABC . It has some interesting properties. For example, it can be shown that among all triangles inscribed in $\triangle ABC$, it has the least perimeter. This was proved by the mathematician **Fagnano**.

There are numerous centres associated with a triangle. The most familiar of them are the circumcentre, the incentre, the centroid and the orthocentre. There are some relationships between them, for example the collinearity of the orthocentre, the centroid and the circumcentre of a triangle. (The line containing them is called the **euler line** of the triangle. This line also contains the centre of the nine-point circle of the triangle, yet another popular centre associated with a triangle.)

But more interesting are the relationships between a particular centre of one triangle and some other centre of another related triangle. And occasionally, they are useful in some proofs. For example, it is easy to prove the concurrency of the the perpendicular bisectors of the three sides of a triangle because of the characterisation that such a side bisector is precisely the locus of a point which is equidistant from the two end points of the side. Proving the concurrency of the three altitudes of a triangle ABC by a similar argument is not possible because there is no similar characterisation of an altitude. (There are, of course, other proofs. There is a pure geometry proof using cyclicity of the quadrilaterals $OFAE$ and $OECD$, which was also used in the solution above. Besides there are proofs based on coordinates and on vectors. See Comments No. 3, 4 and 5 of Chapter 8.)

But using the concurrency of the perpendicular bisectors of another triangle $A'B'C'$, the job can be done. This triangle is obtained by drawing lines through A, B, C which are parallel to the sides BC, CA, AB respectively as shown in the figure below. Clearly, A, B, C are the mid-points of the sides $B'C', C'A', A'B'$ respectively. Moreover these sides are parallel to the sides BC, CA, AB of the original triangle. It follows then that the altitudes of $\triangle ABC$ (shown by dotted lines) are precisely the perpendicular bisectors of the sides of the triangle $A'B'C'$. Hence concurrency of the latter implies that of the former.



A similar relationship between some centre of one triangle and a different kind of a centre of a related triangle gives a quick solution to the present problem. The relationship is that the orthocentre of any triangle is also the incentre of the triangle whose vertices are the feet of the three altitudes of the original triangle. In the notations of the figure in the statement of the problem, O is the incentre of the triangle DEF . If you invoke this result, there is nothing left to prove. For, OE is the angle bisector of $\angle FED$ and so a ray coming along FE will go along ED and after reaching D , will get reflected to move along DF . At F it will get reflected and will start moving again along FE .

It may be argued that those who are familiar with this result get an unfair advantage over those who are not. This is analogous to the situation in Q. 2 above where those who know (and can recall) identity (1) stand to gain. In Q. 3, too, familiarity with the JEE 2001 question gives an advantage. But these can hardly be called unfair gains. What is wrong if knowledge is rewarded?

It may also be argued that the problem asked merely to show the existence of a triangular periodic path. But instead, we have merely *verified* that a particular path has the desired property without giving any construction for this path (other than indicating that the altitudes of the ambient triangle may have a role.)

Logically, the solution is unassailable. A possible reason for skepticism might be the so called **existence theorems**. These are theorems which guarantee the existence of something without giving any explicit

construction for it. Many important theorems in calculus such as the mean value theorems or the Intermediate Value Property (IVP) are of this type. This happens because all these theorems invoke the basis of calculus, called **completeness** of the real number system. (For more elaboration, see my book *Calculus for Scientists and Engineers - An Analytic Approach*, a Narosa publication.)

Sometimes, the use of completeness is inevitable to prove the existence of something. For example, to prove that the equation $\tan x = x$ has a solution in the interval $(\pi/2, 3\pi/2)$ we need the IVP because we cannot explicitly identify any solution. But if the interval is $(-\pi/2, \pi/2)$, then we can explicitly verify that $x = 0$ is a solution. IVP can, of course, be applied here too. But it is not vital.

In the solution to Q. 1 above, we only proved the existence of a fixed point x_0 of the function f . We did give a construction for it, viz. the limit of the sequence obtained by applying iterates of the function f starting from any arbitrary point a_0 . But this is not a finitistic construction. The existence of the limit of every monotonically increasing sequence which is bounded above is one of the formulations of the completeness of the real number system. Moreover the fixed point there was not unique.

One can only guess what might have prompted the paper-setters to underplay by asking only the existence of a certain triangle. One possible answer is that had they asked to show that the orthic triangle has the desired property, the problem would be too simple, little more than an exercise in school geometry. In questions like this the real challenge is to *think of* the right answer. In Q. 2 above, too, the key idea was to think of the right identity (1). But that was not much of a challenge because naturally one would look for an identity for the sum of the squares of the sines of the angles of a triangle.

- Q. 8 Let $n \geq 2$ and let $a_1 \leq a_2 \leq \dots \leq a_n$ be positive integers such that $\sum_{i=1}^n a_i = \prod_{i=1}^n a_i$. Prove that $\sum_{i=1}^n a_i \leq 2n$ and determine when equality holds.

Solution and Comments: $2 + 2$ is probably the first addition a child learns. And possibly for the same reason it is cited, often derogatively,

to describe something as childishly simple. Thus people often say ‘It’s not as simple as two plus two equals four’. And an argument frequently made against mathematics is ‘In mathematics 2 plus 2 always equals 4. But here we are dealing with reality, which is far more complex’.

But indeed there is something very special about the sum $2 + 2$. And the present problem is precisely about this special feature. 4 is not only the sum but also the product of 2 and 2. Normally, while dealing with positive integers, we expect the product to be greater than the sum. If x and y are real numbers, then the equation

$$x + y = xy \tag{1}$$

has infinitely many solutions. If we rewrite this equation as

$$(x - 1)(y - 1) = 1 \tag{2}$$

we see that its solutions are precisely the points on a hyperbola with centre at $(1, 1)$ and asymptotes as the lines $x = 1$ and $y = 1$.

So, there is no dearth of solutions to either (1) or (2) if x, y are allowed to be real numbers. But see what happens if they are restricted to be positive integers. Now $x - 1$ and $y - 1$ are non-negative integers and since the only factorisation of 1 in non-negative integers is 1×1 , from (2) we get that both x and y equal 2. The same reasoning extends to showing that the product of any three (or more) integers is greater than its sum, provided they are all bigger than 1. In fact the difference will get bigger as the number and sizes of these integers grow.

In the present problem, let us call the sum $\sum_{i=1}^n a_i$ and the product $\prod_{i=1}^n a_i$ as s and p respectively. Normally, we expect that $s < p$. But we are given that $s = p$. The only way this is possible is if some of the a_i ’s equal 1 each because 1 as a factor does not increase p but, as a term, it increases the sum s by 1 and so an adequate number of terms in the sequence must be 1 so as to make up the deficit $p - s$. As the sequence is monotonically increasing, these terms must appear at the start, followed by the bigger terms in ascending order.

To be specific, assume $a_1 = a_2 = \dots = a_r = 1$ and $a_{r+1} \geq 2$. Let $p_1 = \prod_{i=r+1}^n a_i$ and $s_1 = \sum_{i=r+1}^n a_i$. Then p_1 is the same as p . But $s = s_1 + r$.

Hence

$$r = s - s_1 = p - s_1 = p_1 - s_1 \quad (3)$$

For notational simplicity, call $n - r$ as k (k is the number of terms in the sequence which are bigger than 1). Also denote a_{r+i} by b_i for $i = 1, 2, \dots, k$. Then

$$n = r + k = b_1 b_2 \dots b_k - (b_1 + b_2 + \dots b_k) + k \quad (4)$$

Since $s = p = p_1$, our goal, viz. to prove the inequality $s \leq 2n$ is equivalent to showing that

$$b_1 b_2 \dots b_k \leq 2(b_1 b_2 \dots b_k) - 2(b_1 + b_2 + \dots + b_k) + 2k \quad (5)$$

which further reduces to proving that

$$b_1 b_2 \dots b_k \geq 2(b_1 + b_2 + \dots + b_k - k) \quad (6)$$

Further let $b_i = 1 + c_i$ for $i = 1, 2, \dots, k$. Then each $c_i \geq 1$ and (6) further reduces to proving that

$$(1 + c_1)(1 + c_2) \dots (1 + c_k) \geq 2(c_1 + c_2 + \dots + c_k) \quad (7)$$

On expansion, the L.H.S. is a sum of $k+1$ terms $T_0, T_1, T_2, \dots, T_k$ where

$$\begin{aligned} T_0 &= 1 \\ T_1 &= \sum_i c_i \\ T_2 &= \sum_{i < j} c_i c_j \\ T_3 &= \sum_{i < j < t} c_i c_j c_t \\ &\vdots \\ &\vdots \end{aligned}$$

$$\text{and finally } T_k = c_1 c_2 \dots c_k$$

In general, T_s is a sum of $\binom{k}{s}$ terms, for $0 \leq s \leq k$. T_1 equals $c_1 + c_2 + \dots c_k$ which is half the R.H.S. of (7). The other half can be taken care of by T_2 provided $k > 2$. For, in that case, included in T_2 are

all products $c_1c_2, c_2c_3, \dots, c_ic_{i+1}, \dots, c_{k-1}c_k$ and c_1c_k . (Note that c_kc_1 is not allowed because of the restriction on the indices. But we can take c_1c_k .) Since all the c 's are at least 1, these products are at least $c_1, c_2, \dots, c_{k-1}$ and c_k respectively. Therefore $T_2 \geq c_1 + c_2 + \dots + c_k$. All put together we have proved the inequality $s \leq 2n$ at least in the case when $k \geq 3$. Actually, we have proved strict inequality because the very first term T_0 was never used.

What about the cases $k = 1, 2$? The case $k = 1$ is degenerate because in this case only the last term of the sequence, viz. a_n is greater than 1. The first $n - 1$ terms equal 1 each. Hence $p = a_n$ while $s = (n - 1) + a_n$. The equality of p and s gives $n - 1 = 0$, i.e. $n = 1$, contradicting the stipulation $n \geq 2$ given in the question itself.

But the case $k = 2$ is interesting. Here $a_1 = a_2 = \dots = a_{n-2} = 1$. In terms of the notations introduced above, $a_{n-1} = b_1 = 1 + c_1$ and $a_n = b_2 + 1 + c_2$ where c_1, c_2 are positive integers. By a direct calculation, $s = n - 2 + b_1 + b_2 = n + c_1 + c_2$ and $p = b_1b_2 = (1 + c_1)(1 + c_2)$. Equating s with p gives

$$n = (1 + c_1)(1 + c_2) - c_1 - c_2 = 1 + c_1c_2 \quad (8)$$

The inequality to be proved, viz. $s \leq 2n$ reduces to proving $n + c_1 + c_2 \leq 2n$, i.e. $n \geq c_1 + c_2$, or equivalently, to proving $1 + c_1c_2 \geq c_1 + c_2$. But this is true because $1 + c_1c_2 - c_1 - c_2$ factors as $(c_1 - 1)(c_2 - 1)$ and both the factors are non-negative integers. This also shows that equality will hold (i.e. $s = 2n$) if and only if at least one of the factors vanishes, i.e. when either $c_1 = 1$ or $c_2 = 1$ (or both). But since both are positive integers and $c_1 \leq c_2$, we must always have $c_1 = 1$. That gives $a_{n-1} = b_1 = 1 + c_1 = 2$, and a_n to be any integer greater than or equal to 2. In the simplest case, c_2 also equals 1 and hence $n = 1 + c_2 = 2$. Thus the sequence has only two terms. viz. 2 and 2. This is the magical case where the sum as well as the product of two numbers are the same. If we give the next higher value to c_2 , viz. $c_2 = 2$, then we get the sequence with three terms, viz. 1, 2, 3. This is also a well-known example, even for a layman, where the sum of three integers is also their product. If we put $c_2 = 5$, we get the sequence 1, 1, 1, 2, 5. Note that in all these examples, $a_n = n$. This is not surprising because $2a_n$ equals p and hence s which equals $2n$.

We already showed that in presence of equality of p and s , equality of s and $2n$ cannot hold for $k > 2$. So a complete answer to the second part of the question is that in a monotonically increasing sequence $a_1, a_2, \dots, a_n$ of positive integers in which the sum of the terms equals their product, the sum equals $2n$ if and only if $a_{n-1} = 2, a_n = n$ and all other terms are 1. ■

An unusual but excellent problem on inequalities.

CONCLUDING REMARKS

After the JEE became totally objective type in 2006, one eagerly looks at the second paper of ISI B.Sc. Entrance Test for some good problems. Usually, every year Paper 2 is a mixed bag which is inevitable since all questions cannot be expected to have the same qualities. But this year, there is a staggering gap between their difficulty levels. The first question is purely analysis and rewards those who go more for analytical proofs than computational prowess. To some extent Q. 3 in Paper 2 is also of this type, even though it relies heavily on a trick adopted to find integrating factors for linear differential equations.

After that there are extremes. Q. 2 and Q. 7 depend on applying certain other results. Once you recollect them correctly, hardly anything is left. But Q. 4 and Q. 6 require long solutions and are more suited as take home assignments or term projects. Q. 5 is a tricky but reasonable question on roots of a cubic. The best question in Paper 2 is Q. 8, characterising when sums of integers equal their products.

Paper 1 also contains some good problems. The best question is Q.30 which rewards those who can do some out of the box thinking. Q.19 which asks for a definite expression for a function defined by an integral is also very good. Q.11 about recurrence relations is also good because the answer comes more easily without solving it. Q.17 is on local extrema, but requires a different kind of thinking than usual. All geometry problems (Q.3, Q.14) are easy. So is the case with combinatorics problems.

Q.24 about the largest equilateral triangle inside a unit square is tricky as it is more a test of the ability to seek short cuts. In the question on

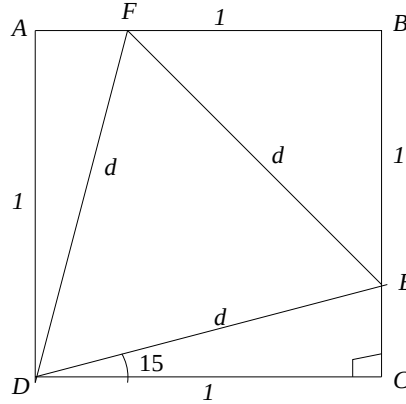
orthic triangles in Paper 2, the answer is not likely to be known beforehand unless you happen to know it beforehand, possibly out of curiosity or random fiddling with trigonometric optimisation problems. While such curiosity is definitely an asset academically, it is clearly unfair to make the solution so crucially dependent on a particular piece of knowledge, not covered in standard syllabi.

As it always happens, the MCQ format, especially the one adopted by ISI where it is given that only one option is correct, has marred some otherwise good problems. A glaring example of this is Q.9 on Pythagorean triples.

LARGEST EQUILATERAL TRIANGLE

Here is an outline of a proof that the side length of a largest equilateral triangle that can be drawn inside a unit square (with edges included) is $\sec 15^\circ$ i.e. $\sqrt{6} - \sqrt{2}$.

The actual construction of such a triangle is easy. Suppose $ABCD$ is a unit square. Through one vertex, say D , draw lines DE and DF each making an angle 15° with the sides DC and DA respectively. Then the triangle with vertices D, E, F is equilateral since $DE = DF$ by symmetry about the diagonal BD and $\angle EDF = 90 - 15 - 15 = 60^\circ$. Also from the right angled triangle DCE , $DE = \sec 15^\circ$.



The value of $\cos 15^\circ$ is not so common. (Some old books do give it.) But we can get it by solving the quadratic $2u^2 = 1 + \cos 30^\circ = 1 + \frac{\sqrt{3}}{2}$ where $u = \cos 15^\circ$. Its reciprocal, $\sec 15^\circ$ comes out to be $\sqrt{6} - \sqrt{2}$. (Moreover, since an angle of 15° is constructible with straight edge and compass, so is the triangle DEF .)

The more challenging part is to show that no equilateral triangle PQR with vertices in $ABCD$ can have its side length exceeding $\sec 15^\circ$. On the face of it this is a problem of non-linear constrained optimisation. For, suppose the vertices are $p = (x_1, y_1), Q = (x_2, y_2), R = (x_3, y_3)$. Then we have to

maximise the objective function

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \quad (1)$$

subject to the three equality constraints

$$(x_1 - x_2)^2 + (y_1 - y_2)^2 = (x_1 - x_3)^2 + (y_1 - y_3)^2 \quad (2)$$

$$(x_2 - x_3)^2 + (y_2 - y_3)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 \quad (3)$$

$$\text{and } (x_3 - x_1)^2 + (y_3 - y_1)^2 = (x_3 - x_2)^2 + (y_3 - y_2)^2 \quad (4)$$

(actually any two of these are sufficient) and the twelve inequality constraints

$$0 \leq x_i \leq 1, 0 \leq y_i \leq 1 \quad i = 1, 2, 3 \quad (5)$$

The constraints (2)-(4) can be simplified a little. Still they will remain non-linear. So the methods of linear programming cannot be applied here. The calculus method of maximisation based on identifying the critical points is also not practicable directly because of the large number of variables and the complicated expressions for the partial derivatives.

But we can implement the essence of the calculus method in a different manner. In essence, a critical point of a sufficiently smooth function of several real variables is a point, say P , at which all partial derivatives, and hence all directional derivatives of the objective function vanish. That means no matter in which direction you move from P , the function will decrease (or increase, depending upon whether we are trying to maximise or minimise it), provided the displacement is infinitesimally small (or **marginal**, to use a less technical word).

The contrapositive of this approach is that if P is a candidate solution for a maximum point of some function and if we can find some direction in which the function increases at P , then definitely, P is not a right candidate for maximisation. If we move slightly in this direction, we shall reach a point Q which is a better candidate than P . It may happen, however, that at Q , there is some other direction in which the function will increase and that will yield to a still better solution. Since only infinitesimally small displacements are occurring at each improvement the direction of the path may change at every point. So the actual path traced may not necessarily be straight line path but some curve from P to Q with the property that the function is increasing in the direction of the tangent to it at each point.

So, the general plan of the proof is to start with an arbitrary equilateral triangle PQR whose vertices all lie in the unit square. We shall show PQR can be changed to another equilateral triangle $P'Q'R'$ whose side is longer than that of PQR . If no such triangle exists, the $\triangle PQR$ is the desired largest triangle. Otherwise we apply this procedure to $\triangle P'Q'R'$ to get another equilateral triangle $P''Q''R''$ which has a larger side. And this process is repeated until we reach an equilateral triangle whose side is $\sec 15^\circ$ and for which no improvement is possible. That will prove that the maximum possible side length is $\sec 15^\circ$.

There are infinitely many possible equilateral triangles to start with. To carry out this journey systematically, we start with the set, say $\mathcal{S}_0$ of all equilateral triangles inside the unit square and gradually reduce the subclass of it consisting of candidates for maximality to smaller and smaller classes. The definitions of these classes will be in terms of the possible locations of their vertices.

We list these classes.

- (0) $\mathcal{S}_0$ all possible equilateral triangles with vertices in the unit square
- (II) $\mathcal{S}_1$ all equilateral triangles with at least one vertex on a side of the square
- (II) $\mathcal{S}_2$ all equilateral triangles in $\mathcal{S}_1$ with at least two vertices on sides of the square
- (III) $\mathcal{S}_3$ all equilateral triangles in $\mathcal{S}_3$ with all three vertices on the sides of the square
- (IV) $\mathcal{S}_4$ all triangles in $\mathcal{S}_3$ with one vertex at a corner of the square.

It is not claimed, nor is it true that all triangles in each class have the same size i.e. the same side length. In fact, this is patently false for the class $\mathcal{S}_0$ where the side could be any number from 0 (the degenerate triangle with only one point in it) to the maximum we are seeking. The purpose of classification is different. Let PQR be an equilateral triangle for which d , the side length, is largest. The existence of such a triangle, without exhibiting it, follows from calculus by noting that the function d defined by (1), is a continuous real valued function of four real variables taking values in the unit interval, which is a closed and bounded set. We shall progressively narrow

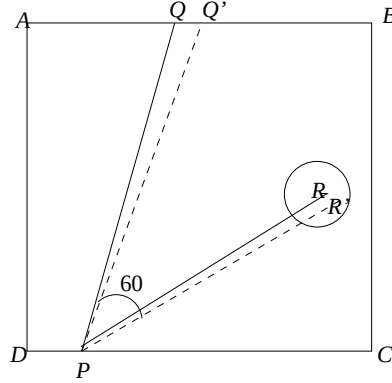
the search for it from $\mathcal{S}_0$ to the class $\mathcal{S}_4$ and the classes listed above are the intermediate stops on the way. Once we arrive at $\mathcal{S}_4$, it will be easy to show that $\triangle PQR$ has only four possibilities all of which are congruent to the triangle FDE in the figure above.

So our job is to show that if the largest equilateral triangle PQR is in class $\mathcal{S}_i$ for some $i = 0, 1, 2, 3$, then we can find an equilateral triangle $P'Q'R'$ in the class $\mathcal{S}_{i+1}$ whose side length is at least that of $\triangle PQR$. We carry these out one-by-one. In each transition except the last one, we shall follow the method of marginal increment.

- (1) **From $\mathcal{S}_0$ to $\mathcal{S}_1$** : If the triangle PQR is not in $\mathcal{S}_1$, then all its vertices P, Q, R are interior points of the unit square. So there exists some $\delta > 0$ such that discs of radii of δ centred at P, Q, R will all lie inside the square $ABCD$. So, we can dilate $\triangle PQR$ to a bigger triangle $\triangle P'Q'R'$ which is also equilateral, contained in the square. (A specific construction for this dilation is to consider the centre, say M , of $\triangle PQR$ and to extend the segments MP, MQ, MR equally to get bigger segments MP', MQ', MR' respectively.) Then $\triangle P'Q'R'$ is similar to but larger than $\triangle PQR$, contradicting the maximality of the latter.
- (2) **From $\mathcal{S}_1$ to $\mathcal{S}_2$** : We now know that at least one vertex, say P , of the largest triangle PQR lies on some side say AB , of the unit square $ABCD$. Suppose neither of the remaining two vertices Q and R lies on any side of the square. Then both are interior points and hence for some sufficiently small $\delta > 0$, the open discs of radii δ centred at Q and R are entirely contained in the square. Elongate PQ, PR equally to PQ' and PR' respectively so that Q', R' are within these discs and hence in the unit square. Then as in (i), $\triangle P'Q'R'$ is similar to $\triangle PQR$ and hence is equilateral, but has side greater than that of the latter, a contradiction. So, at least one out of Q and R lies on the boundary of the square.
- (3) **From $\mathcal{S}_2$ to $\mathcal{S}_3$** . Now we know that at least two of the vertices, say P and Q , of the largest equilateral triangle PQR lie on two sides of $ABCD$. This side cannot be common to both. For if it were then the side of the triangle will be at most 1. But we can easily draw equilateral triangles inside $ABCD$ whose sides are bigger than 1. (Actually, in the figure above, $\triangle DEF$ is one such triangle since $\sec 15^\circ > 1$.)

So, P and Q lie on different sides of the square $ABCD$. We have to consider two cases depending upon whether these two sides are opposite or adjacent. In both the cases, the third vertex R will be an interior point (as otherwise $\triangle PQR$ is already in class $\mathcal{S}_3$. Hence R can move inside an open disc of radius δ for some $\delta > 0$. We shall use this freedom of motion to construct a bigger equilateral triangle inside $ABCD$.

Assume first that P and Q lie on opposite sides, say AB and CD of the unit square. Keep P fixed and move Q slightly to a point Q' on AB which is farther from P than Q is as shown in the figure below.



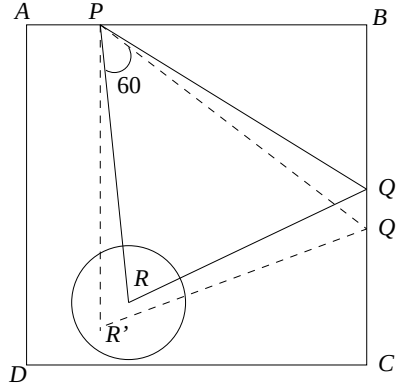
Then $PQ' > PQ$. The angle QPR is 60° . But $\angle Q'PR$ is slightly smaller. But we can move R slightly to a nearby point R' so that $\angle Q'PR' = 60$ degrees and $PR' = PQ'$. (In effect, R' is obtained from R by first rotating PR through the angle $\angle QPQ'$ and then elongating it by the factor $\frac{PQ'}{PQ}$. The angle $\angle QPQ'$ has infinitesimally small measure

because QQ' is infinitesimally small. Moreover, the ratio $\frac{OQ'}{OQ}$ exceeds 1 by an infinitesimally small number. Hence when these infinitesimals are sufficiently small, R' is inside the disc centred at R and therefore is also inside the square $ABCD$.)

The new triangle $Q'PR'$ is an equilateral triangle with all three vertices lying inside the square $ABCD$. But its side PQ' is bigger than the side PQ of $\triangle PQR$, contradicting the assumed maximality of $\triangle PQR$. (This construction fails if Q is the end-point B of the side AB . In that case, hold Q fixed and move P on CD so that it moves farther from Q . If

this also fails then P coincides with D . But that would mean that PQ is a diagonal of the square and obviously there cannot be an equilateral triangle inside $ABCD$ with side $\sqrt{2}$.)

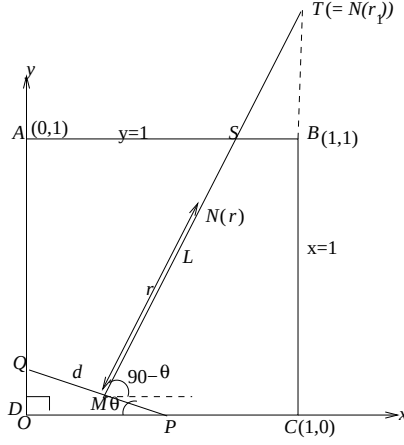
The other case, where two vertices P and Q lie on adjacent sides, say, AB and BC respectively of the unit square $ABCD$ and R is an interior point is handled similarly. Move Q (or P) so that the distance PQ increases slightly. Then rotate PR and also elongate it to get an equilateral triangle $P'Q'R'$ which is bigger than $\triangle PQR$.



The only way this approach will fail is when P and Q are already at the corners A and C respectively. But then $\triangle PQR$ will contain two diagonal vertices, a contradiction to its equilaterality.

Summing up, if a largest equilateral triangle PQR has at least two vertices on the sides, then it has all three vertices on the sides.

- (4) **From $\mathcal{S}_3$ to $\mathcal{S}_4$.** In this last phase, the possible class in which the assumed largest equilateral triangle PQR lies is already narrowed down to $\mathcal{S}_3$, i.e. P, Q, R all lie on some edges of the unit square $ABCD$. At least two of these edges, say CD and AD , are adjacent. Assume they contain P and Q respectively. Then the third vertex R must lie on one of the other two sides. Also R lies on the perpendicular bisector, say L , of the segment PQ . Let M be the midpoint of PQ . Let S and T be the points of intersection where L meets the lines AB and BC respectively. By assumption R is S or T .



Introduce coordinates with D as the origin O and DC and DA as the x - and y -axes respectively. Let θ be the angle QPD . Then $P = (d \cos \theta, 0)$, $Q = (0, \cos \theta)$ and $M = (\frac{d \cos \theta}{2}, \frac{d \sin \theta}{2})$. Let $d = PQ$. That is, d is the side length of the equilateral triangle PQR . To get the coordinates of R , we use the fact that R lies on L at a distance $\frac{\sqrt{3}d}{2}$ from the point M . So we first get a parametrisation of the line L (or rather, the half of it lying on the other side of O) in terms of the distance, say r , of a point on it from M . Since L is perpendicular to PQ , the angle it makes with the horizontal direction (shown by a dotted line) is $\frac{\pi}{2} - \theta$. Hence a unit vector in the direction of L is $(\sin \theta, \cos \theta)$. So, if N is a point on L , at a distance r from M , then

$$N = N(r) = \left(\frac{d \cos \theta}{2} + r \sin \theta, \frac{d \sin \theta}{2} + r \cos \theta \right) \quad (1)$$

Separately, the x - and y -coordinates of $N(r)$ can be expressed as functions of r by

$$x(r) = \frac{d \cos \theta}{2} + r \sin \theta \quad (2)$$

$$\text{and } y(r) = \frac{d \sin \theta}{2} + r \cos \theta \quad (3)$$

In (1), d and θ are fixed and r is the parameter. The point R results by taking $r = \frac{\sqrt{3}d}{2}$. After some simplification it comes out as

$$R = N\left(\frac{\sqrt{3}d}{2}\right) = \left(d \sin\left(\theta + \frac{\pi}{6}\right), d \sin\left(\theta + \frac{\pi}{3}\right) \right) \quad (4)$$

It is quite obvious that if $\theta = \frac{\pi}{4}$, then by symmetry about the line $x = y$, L will pass through the vertex B diagonally opposite to D . However, the value of r for which this happens will depend on d .

We claim that for $\theta \neq \frac{\pi}{4}$, the points S and T are distinct and neither of them equals R . Note that the slope of L is $\cot \theta$. For motion along L , the rate of growth of r horizontally is $\sin \theta$ and that along the vertical is $\cos \theta$.

From (2) and (3), both $x(r)$ and $y(r)$ are strictly increasing functions of r . Let r_1 be the unique value of r for which $x(r) = 1$. Then r_1 is given by $\frac{d \cos \theta}{2} + r_1 \sin \theta = 1$, which upon solving gives

$$r_1 = \frac{2 - d \cos \theta}{2 \sin \theta} \quad (5)$$

$N(r_1)$ is the point T where the line L meets the line $x = 1$, i.e. the line containing the edge BC of the square $ABCD$.

By an entirely similar argument, the point S at which L cuts the line $y = 1$ is $N(r_2)$ where

$$r_2 = \frac{2 - d \sin \theta}{2 \cos \theta} \quad (6)$$

Assume first that $0 < \theta < \frac{\pi}{4}$. We compare r_1 and r_2 . If $r_1 > r_2$, then starting from the point M , on the line L , the point S comes before T does. (The case illustrated in the diagram is of this type.) S is on the edge AB and as observed above, R , the third vertex of the equilateral triangle PQR has to equal S by our assumption that all the vertices of $\triangle PQR$ lie on the sides of the square $ABCD$.

The comparison between r_1 and r_2 can be done graphically by preparing two columns headed by r_1 and r_2 and the first row containing the values of r_1 and r_2 given by (5) and (6). In each row after the first, we reduce the comparison by applying some order preserving operation to the entries of the previous row (e.g. multiplying both the entries by a positive constant or adding/subtracting the same number from both the sides).

r_1	r_2
$1 - \frac{d \frac{\cos \theta}{2}}{\sin \theta}$	$1 - \frac{d \frac{\sin \theta}{2}}{\cos \theta}$
$\cos \theta (2 - d \cos \theta)$	$\sin \theta (2 - d \sin \theta)$
$2 \cos \theta - d \cos^2 \theta$	$2 \sin \theta - d \sin^2 \theta$
$2(\cos \theta - \sin \theta)$	$d(\cos^2 \theta - \sin^2 \theta)$
2	$d(\sin \theta + \cos \theta)$

In the first reduction we have used that $\sin \theta$ and $\cos \theta$ are both positive while in the last one, we have used $\cos \theta > \sin \theta$ which holds since $0 < \theta < \frac{\pi}{4}$.

The last comparison shows that $r_1 > r_2$ because both $\sin \theta + \cos \theta$ and d are less than $\sqrt{2}$ (by our assumption that $\theta \neq \frac{\pi}{4}$) and because only the diagonal vertices can be maximally apart in a square. (If P and Q are diagonally apart then R will be outside the square.)

When $\frac{\pi}{4} < \theta$, the tables are turned around and we get $r_1 < r_2$, which means on L , starting from M , T comes before S and so R must equal T .

Whether in the case $0 < \theta < \frac{\pi}{4}$, or the case $\frac{\pi}{4} < \theta < \frac{\pi}{2}$, so far, the maximality of $\triangle PQR$ was not used. (We did use it, of course, in the earlier transitions, that is, from $\mathcal{S}_0$ to $\mathcal{S}_1$, then from $\mathcal{S}_1$ to $\mathcal{S}_2$, then from $\mathcal{S}_2$ to $\mathcal{S}_3$ and finally from $\mathcal{S}_3$ to $\mathcal{S}_4$.) We now show that under the additional assumption of maximality, one of the vertices must be a corner. (We already showed this in the case when $\theta = \frac{\pi}{4}$.)

So, once again, without loss of generality assume that $0 < \theta < \frac{\pi}{4}$. We claim that if $\triangle PQR$ has the largest possible side d among all equilateral triangles that can be drawn inside the unit square $ABCD$, then the vertex P must be the corner C and further, d must equal $\sec 15^\circ = \sqrt{6} - \sqrt{2}$. (At the start of this article, we already constructed onw such triangle with these specifications. But that time we did not prove its maximality. We are now in a position to do that.)

We already showed that when $0 < \theta < \frac{\pi}{4}$, the third vertex R must coincide with the point S where the line L meets the side AB . The y -coordinate of S is 1. Equating it with the y -coordinate of R as given

by (4), and using some elementary trigonometric simplification, we get

$$\begin{aligned}
d &= \operatorname{cosec}\left(\frac{\pi}{3} + \theta\right) \\
&= \sec\left(\frac{\pi}{2} - \frac{\pi}{3} - \theta\right) \\
&= \sec\left(\frac{\pi}{6} - \theta\right) \\
&= \sec\left(\left|\frac{\pi}{6} - \theta\right|\right)
\end{aligned} \tag{7}$$

where, in the last step we have used that $\cos \alpha = \cos(-\alpha) = \cos(|\alpha|)$ for any α .

Maximising d now reduces to maximising $\sec(|\frac{\pi}{6} - \theta|)$ and hence to maximising $|\frac{\pi}{6} - \theta|$ (because over the interval $[0, \frac{\pi}{4})$ and even over the bigger interval $[0, \frac{\pi}{2})$, the secant function is strictly increasing).

By an elementary argument as θ varies over $(0, \frac{\pi}{4})$, $|\frac{\pi}{6} - \theta|$ is maximum at $\theta = \frac{\pi}{12}$ and again at $\theta = \frac{\pi}{4}$. But the latter is an end point of $(0, \frac{\pi}{4})$ and not in the domain.

As observed above, the case $\frac{\pi}{2} < \theta < \frac{\pi}{2}$ can be dismissed using symmetry of the functions about the midpoint $\frac{\pi}{4}$. The maximum of d will occur when $\theta = \frac{5\pi}{12} = 75^\circ$.

Thus we have shown that any largest equilateral triangle has side length $\sec 15^\circ = \sqrt{6} - \sqrt{2}$, one vertex at one of the corners of the square and is symmetric about the diagonal passing through this corner. We have already constructed one such triangles. Its rotations through multiples of $\frac{\pi}{2}$ give, in all, four largest triangles.